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A practical approach on scenario generation & reduction algorithms for wind power forecast error scenarios

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Abstract

Abstract goes here.

1 Introduction

Present-day and future power systems are rife with uncertainty, mainly due to the integration of intermittent renewables. Availability and output of renewable generators can fluctuate rapidly with changes in weather. Novel system operation methods, such as stochastic unit commitment models (SUCMs), will be needed to ensure a safe and cost-efficient operation of the power system. In low carbon power systems with high shares of renewables (RES), ensuring reliability will become an increasingly critical issue. Some forms of RES, notably wind and solar PV, have a stochastic character, i.e., they are variable (not or only limitedly dispatchable) and to some extent unpredictable. When forecast errors are made, deviations from what was expected need to be overcome with up- or downward regulation of dispatchable generation, load or storage. Moreover, the variability of these RES injections requires this reserve capacity to be sufficiently flexible. In this regard, novel power system operation methods will be needed to properly size and allocate operational reserves, in order to ensure a safe and cost-efficient operation of the power system.

Most unit commitment models however are deterministic in nature. A typical approach to handling uncertainty in deterministic unit commitment models (DUCMs) is to build in extra capacity in to the schedule, known as reserves, to allow the power system to accommodate unexpected changes. In SUCMs, the uncertainty is represented via a number of realizations of that uncertainty, i.e. scenarios. The problem is then typically represented as a two-stage recourse problem. One tries to find a unit commitment as such that a feasible dispatch is possible for all possible realizations (scenarios) of an uncertain variable, in this case the WPFE. A feasible dispatch here means that the demand for electricity in each time step is met in all of the considered WPFE scenarios, while respecting all techno-economic constraints of the power plants. In such a stochastic model, the so-called *here-and-now* or *first stage* variable – i.e. the optimization variable(s) that common to all scenarios – is thus the unit commitment status. All other optimization variables, such as the output of each power plant, are the so-called *wait-and-see* or *second stage* variables – i.e. the variable(s) that take on different values in each scenario. Such models may offer a number of advantages over DUCMs by the way they treat the uncertainty (e.g. see [1]). However, the added value of these models is fully dependent on the quality of the scenarios that one uses in this model. These scenarios are often regarded as input data, but in reality, they are a critical part of your model [2]. In Section 5 we will illustrate that without the 'best' possible set of scenarios, the stochastic model will not find an optimal solution.

Often one starts from a continuous description of the stochastic variable. Solving a unit commitment model with such a continuous representation of a stochastic variable is however next to impossible. Modelers then resort to **scenario generation** algorithms to translate the continuous description of the stochastic variable to a set of discrete realizations of that stochastic variable, i.e. scenarios. However, the computational burden of solving such a SUCM with a large number of scenarios is extremely high and for practical purposes often impossible. To avoid so-called intractability, **scenario reduction** algorithms are used to limit the number of scenarios needed in the stochastic model. These algorithms are designed to ex-ante select those scenarios that affect the value of the first-stage variables, and thus the quality of the solution, of the stochastic program. The challenge here lies in the ex-ante character of the selection: one needs to decide which scenarios will affect the solution of the problem, before solving the problem. In the next section, we will detail these requirements for scenario reduction algorithms and refer to these requirements as stability and bias. Modelers thus face the challenge of designing a set of representative scenarios that captures all significant realizations of the stochastic variable while limiting the computational cost of solving the resulting stochastic program, without loss of quality of the solution.

In this paper, we will give an overview of some theoretical elements of scenario generation and reduction, as well as some requirements for what we believe to be a good scenario generation and reduction technique (Section 2). The focus is on the practical application, not the theory, for which an interested reader is referred to specialized literature (e.g. [3]). Throughout the paper, we will focus on the scenario generation and reduction methods for wind power forecast errors (WPFE), currently a very active field of research. Our results and recommendations can however easily be used to model other sources of uncertainty. In Section 3, an overview of scenario generation techniques will be given[1]. We will in detail describe the scenario generation technique that we believe to be adequate to generate a set of scenarios that represents the stochastic character of WPFE. Section 4 treats scenario reduction techniques, with a special focus on the probability distance-based scenario reduction techniques such as the *fast forward* algorithm [4]. Before formulating a conclusion and some suggestions for future work, we will illustrate the added value of these scenario generation and reduction techniques in case study inspired by the Belgian power system. First, we will show that the selected scenario generation technique leads to a relatively small set of scenarios that captures the stochastic behavior of the Belgian WPFE. For this modeling exercise, we will start from the start from the statistical description developed in Bruninx et al. [**?**]. The correspondence of the resulting scenarios and the original distribution will be verified via standard stastical methods, as well as via an event-based verification framework [5]. Second, the added value of the

[1] The provided overview is non-exhaustive. If you feel that some reference is missing, please send it to kenneth.bruninx@kuleuven.be.

scenario reduction technique will be illustrated using a SUCM of the Belgian power system with a (assumed) high penetration of wind power, as described in Bruninx et al. [6].

2 Scenario tree generation & reduction: the basics

Setting up the relevant set of scenarios with a manageable cardinality is one of the most difficult steps in developing a stochastic programming model. Typically, one starts from a large set of historical time series. Given this time series, one tries to understand the underlying random processes governing your model. In the case of wind power forecast errors (WPFE), Bruninx et al. [?] have fitted a conditional Lévy α-stable distribution to the WPFE data of Belgium. Alternatively, one can start directly from knowledge of the underlying random process. Based on this knowledge, one uses a scenario generation technique to generate a number of realizations of the stochastic variable (i.e. a scenario). To adequately represent that stochastic process, one needs to generate a sufficient number of scenarios: the set of scenario needs to cover the most plausible realizations of the stochastic process. This often requires a very large number of scenarios, typically generated via a scenario generation method (see Section 3.1). However, this increases the problem size considerably and may render the optimization problem computationally intractable [3]. Therefore, a scenario generation technique is often combined with a scenario reduction tool (Section 4.1) too reduce the cardinality of the set of scenarios and to regain tractability. As will be illustrated below, the importance of the selected scenarios cannot be underestimated [2, 3, 7, 8]. They are a critical part of your model, not just a part of your data! Furthermore, scenario generation and reduction methods need to be adapted to the optimization problem at hand – no scenario reduction method will fit all purposes.

In this section, we will first elaborate on these scenarios and the so-called scenario trees. The discussion will be general and not applied to the specific case of wind power forecast errors. It is however not the aim of this work to go into detail, but to give some needed background to understand the rest of the paper. Second, we will discuss the governing ideas behind scenario generation and reduction. We will deal with scenario generation and reduction in detail in Sections 3 and 4, where we will apply these ideas to wind power (forecast error) scenarios. Last, a list of requirements for a good scenario tree generation and reduction algorithm are given, to which we will hold the scenario generation and reduction techniques discussed in the next sections.

2.1 Scenario tree: basic concepts

The structure of the optimization problem that the decision maker faces is conveniently visualized through a scenario tree. Visually, a scenario tree consists of nodes and leafs. A node represents the state of a problem at a particular instance – i.e. a the instance at which a decision is made – and are often referred to as stages. The first node is the root node and indicates the beginning of the planning horizon: at this node or stage, the so-called *here-and-now*-decisions are taken. These variables are fixed before the realization of the stochastic process. The root node is connected via branches to the second-stage nodes, where the *wait-and-see* variables are fixed after realization of the stochastic process. The branches or arcs – the paths from the root node to the leafs – are realizations of the random variables, i.e. scenarios. Each branch has a probability of occurrence. The probability of a scenario is thus given by the product of all arc probabilities in that scenario. For a two-stage decision process – like the problem at hand – the second-stage nodes are equal to the scenarios and are called leaves [3]. The probability of a scenario is thus equal to the arc probability. The scenario tree is then characterized by a fan-like structure [2] (see Fig. 1).

2.2 Scenario tree generation & reduction

All scenario generation and reduction techniques have the same goal: minimizing the error caused by the approximation the true stochastic process $\tilde{\epsilon}_t$ with a scenario tree $\breve{\epsilon}_t$ [7, 9]. This approximation error is defined by Pflug [9] for a a two-stage (one-period) stochastic programming model

$$\min_{x\in\mathbf{X}} \; F(\mathbf{x}, \tilde{\boldsymbol{\epsilon}}_\mathbf{t}) \tag{1}$$

as follows:

$$e_f(\tilde{\epsilon}_t, \breve{\epsilon}_t) = F\left(\operatorname{argmin}_{x\in\mathbf{X}} \; F(\mathbf{x}, \breve{\boldsymbol{\epsilon}}_\mathbf{t}); \tilde{\boldsymbol{\epsilon}}_\mathbf{t}\right) - \min_{x\in\mathbf{X}} \; F\left(\mathbf{x}, \tilde{\boldsymbol{\epsilon}}_\mathbf{t}\right). \tag{2}$$

Note that not the optimal solutions $\mathbf{x}$, but their corresponding values of the objective function are compared. The reason is that the objective function of a stochastic programming problem is typically flat, leading to similar objective values for different solutions. The definition (2) is of little practical use, as an evaluation of the objective function with the true stochastic process $\tilde{\epsilon}_t$. If this would be possible, no scenario trees would be needed. However, through simulation and certain approximations, one might be able to test the quality of a scenario generation or reduction method (Section 2.3).

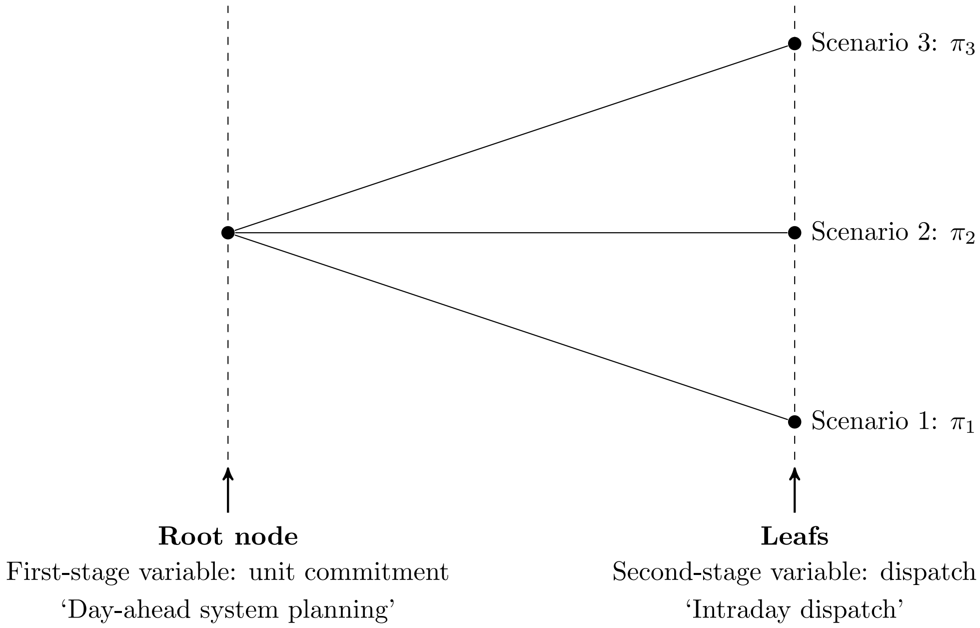


Figure 1: A scenario tree for a two-stage decision problem with tree scenarios with probability π_i. In the problem at hand, at the root node the system planner has to decide on the unit commitment problem ('Day-ahead system planning'). In the second stage, the dispatch is excecuted ('Intraday dispatch').

There are so many techniques as there are no obvious ways to pass from random variables (continuous) to a (good, and what is good in this question) discrete version of those variables [8]. One could seek a discretization which makes the scenario tree as similar to the original distribution as possible, without regard to the stochastic program needing them. However, this will typically lead to very large scenario trees, which might lead to intractable problems, or if you are able to solve the problem at hand, the solution might be no good. In the latter case, increasing the number of scenarios to gain a good result, might then lead to the former case (intractable problem). In conclusion, the quality of a scenario generation or reduction method depends on the model in which it is used. Therefore, scenario generation and/or reduction methods needs to be adapted to the optimization model at hand – no scenario reduction model will fit all purposes. Therefore, before moving on to scenario generation and reduction methods, we need to answer the question *'What is an adequate scenario tree for the problem at hand?'*.

2.3 Requirements for an adequate scenario tree for the wind power forecast error

In essence, a good scenario tree yields the same solution of optimization problem as the stochastic problem if it could be solved with the full, continuous description of stochastic variables. This is off course difficult to test, as it requires to solution of a full stochastic program, but some criteria can be defined [3, 7, 10]

- The **number of stages** and the **number of nodes** should directly reflect the decision process you are studying – in this case two stages is an evident choice. We will study a unit commitment problem with a 24 hour horizon and a 15 minute temporal resolution, so each scenario should contain 96 nodes (values);
- The set of scenarios should **characterize the stochastic variable** as good as possible. The set of scenarios needs to capture all events that affect the unit commitment decision and the probability with which they occur. For the case of wind power forecast errors, this means – amongst others – that the scenarios should respect the heavy-tailed character of the wind power forecast error and its variability;
- The set of scenarios should be as **small** as possible to avoid computational intractability;
- The resulting solution of the stochastic program should exhibit **in- and out-of-sample stability**, indicating that the addition of more scenarios to the optimization does not change the objective function of the stochastic program (too much);
- The solution obtained with the reduced scenario tree should be **unbiased** with respect to the true solution obtained from a stochastic optimization employing a continuous description of the stochastic variable.

From a computational viewpoint, one strives to a small scenario set. The stability-requirement however sets a lower limit on the number of scenario considered in the optimization to obtain meaningful results, as we will illustrate in Section 5. In this paper, we will a propose a two-step approach to obtain a meaningful scenario tree that captures the stochastic variable 'wind power forecast errors'. The first step, a scenario generation method will be employed to generate a limited set of scenarios that form a meaningful representation of the wind power forecast error. Starting from some statistical description of the error (for details, see Bruninx et al. [**?**]) we generate a number of scenarios. The quality of this set is verified by comparing the resulting distributions of the error and its variability, as well as via a event-based verification framework [5]. As such, requirement 2 and 3 are fulfilled. The second step consists of a probability distance-based scenario reduction technique, with with we will try to select those scenarios from the generated set that trigger the optimal unit commitment decisions in the stochastic problem. To test how many scenarios are needed to force our model to take those optimal decisions, we will test the solutions in terms of **stability** (requirement 4) and **bias** (requirement 5):

Stability Kaut and Wallace [7] argue that *stability* can be tested by solving the stochastic program with several different scenario trees of increasing cardinality, generated by the same method. If the objective value does not change (too much), we can claim in-sample stability. The second requirement for stability is out-of-sample stability, which guarantees that the true objective function for the set optimal first-stage decision variables, obtained from solving the stochastic program, does not change (too much) if the cardinality of the scenario tree increases [7]. This can easily be tested by fixing the first stage decision variables –i.e. the unit commitment– to the solution of the SUCM and solving all second stage problems resulting from the original set of scenarios. These stability requirements will set a lower bound on the cardinality of the reduced scenario set.

Bias The *optimality gap* or *bias* of the proposed scenario generation technique should be small. The optimality gap is the difference between the value of the true objective function at the optimal solution of the original problem (with a continuous, full description of the stochastic variable) and the approximation of the problem (with a discrete representation of the stochastic variable – i.e. the scenario tree). As calculating this error would require solving the original problem, approximations have been developed. Kaut and Wallace [7] argue that a stochastic upper bound for the bias $err(\tilde{\mathbf{x}})$ of the method can be obtained as follows, assuming a stable scenario reduction technique:

$$err(\tilde{\mathbf{x}}) \lessapprox \frac{\mathbf{1}}{\mathbf{M}} \sum_{\mathrm{m=1}}^{\mathrm{M}} \left[\mathbf{F}(\tilde{\mathbf{x}}, \breve{\xi}_{\mathbf{m}}) - \min_{\mathbf{x} \in \mathbf{X}} \ \mathbf{F}\left(\mathbf{x}, \breve{\xi}_{\mathbf{m}}\right) \right] \tag{3}$$

with $\tilde{\mathbf{x}}$ the first stage solution based on the selected, stable scenario tree. The set of scenario trees $m \in \{1, 2, \ldots, M\}$ can be chosen freely. As the proposed scenario reduction technique is deterministic, we will perform the test with M trees with slightly different cardinality.

2.4 The structure of the proposed WPFE scenario generation and reduction tool

In the two following sections we will subsequently discuss available scenario generation and reduction techniques. From these techniques, we will propose one to model the wind power forecast error. The selected techniques will be discussed in detail. As will be indicated below, we will propose a separate scenario generation and reduction technique. This will result in a **four-step procedure**. The first step will be the **characterization of the wind power forecast error** [11]. Step two consists of the **generation of a large set of scenarios** that captures the statistical properties of the wind power forecast error. This set serves as an input of step three, i.e. the **scenario reduction**, in which one tries to select those scenarios that will yield a stable and unbiased solution of the stochastic program. **Solving** this **stochastic program**, considering the reduced set of scenarios, is step four. Note that throughout the discussion we assume that a wind power forecast is given, from historical data or calculated from wind speed data, on which we will superimpose a forecast error as generated below.

In the present paper we will discuss step two and three, i.e. the scenario generation and reduction technique. For now, it suffices to know that in [11] we obtained a detailed description of the pdf of the WPFE and the variability of this forecast error. For more information on the stochastic unit commitment problem used later in this paper, the reader is referred to [1, 12].

3 Scenario generation

In this section, we will discuss scenario generation methods, with a specific attention to the application to wind power forecast (error) scenario generation. Note that we will always assume that we have a forecast given (from data or numerical models, for wind power or wind speed which is then transformed to wind power), for which we will then generate 'error scenarios'.

Throughout the discussion, we will have to pay attention to some specific statistical properties of wind power forecast errors, as discussed in Bruninx et al. [11]. First, the wind power forecast is heavy-tailed, meaning that the probability of severe errors is higher than one would obtain under the assumption of a normal distribution. This normality assumption does not hold. As an alternative, we propose the Lévy α-stable distribution [11]. Second, the variability of the error is important and will have to be represented in the forecast error scenarios. Last, the forecast error (cross-)correlated: the error at a certain time step is dependent on the error in the previous and subsequent time steps. All of these properties will have to be reflected in the generated scenarios.

The remainder of this section is organized as follows. First, we will try to give an extensive – but not exhaustive – overview of current scenario generation techniques. For each technique, we will list some advantages and disadvantages. Some examples will be listed, with special attention for applications in the field of wind energy. The overview below is based on Conejo et al. [3] and Dupacova et al. [2]. We will conclude this section a detailed description of the scenario generation technique we have selected, based on Pinson et al. [13] and Ma et al. [14].

3.1 Overview of the scenario tree generation methods

In the literature, one finds an abundance of scenario generation methods. However, no widely accepted classification exists. In this paper, we discuss four categories of scenario generation techniques:

- Sampling
- Property matching
- Path-based methods
- Optimal scenario generation techniques based on probability metrics

Some scenario generation methods do not fit in these three (wide) categories and are not discussed below. Note that some authors list scenario reduction methods as a specific method for scenario generation. In practice, most researchers will however generate some (large) set of scenario, which is later reduced via a scenario reduction method. In the present paper, we will therefore discuss the scenario generation and scenario reduction separately.

Sampling

Principle Given a desired structure of the scenario tree, at every node, we sample several values from a time series or a known underlying distribution. Given the distribution, one can obtain the probability of the sampled value or scenario. Multiple variants of sampling exists, of which random or Monte Carlo sampling is the best-known and simplest form. Samples are randomly drawn from the distribution. If the samples are drawn directly from a time series (i.e. a data sample), the technique is referred to as bootstrap sampling. If sampling can be restricted to a certain range of values, or if sampling can occur according to a certain distribution,. . . this is called importance sampling. Conditional sampling takes into account the relationship with the previous values of the uncertain variables, ensuring the internal consistency of a scenario. Markov chains can be seen as a form of conditional sampling [2].

These methods may be complemented with statistical approaches to check the solution quality during the optimization process. If the solution quality does not meet certain criteria, a larger sample size (tree size) is used to solve the problem once more. These techniques are often referred to as 'internal sampling' or 'importance sampling'. This technique is discussed in Section 4.1.

Pros & cons In all but the simplest of cases, it will be very difficult to randomly sample the 'correct' scenarios: or the tree is too small to represent the underlying distribution (representation problem), or the tree is too large to yield a tractable optimization problem (numerical problem) [8]. Note however that the sampling method has perfect limit properties: if the tree is large enough, it will be a perfect representation of the underlying stochastic process. Furthermore, sampling requires full knowledge of the underlying distribution [7]. Conditional sampling allows to account for the inter temporal relationship of the values of the uncertain variable within a scenario. It is difficult to handle random vectors (multivariate random variables). The size of the tree grows exponentially with the dimension of the random vector if the marginals (univariate components) are sampled separately and recombined via a tensor product. However, several expansions and corrections have been proposed to circumvent these drawbacks. Internal sampling methodologies may not yield convergence for large problems.

Examples In the Sample Average Approximation Method (SAA) by Kleywegt et al. [15], a number of scenario trees of a certain size are sampled from the underlying distribution. If the estimated optimality gap (the approximation error) is sufficiently small, the best solution is retained, otherwise, large trees or more trees of the same size are sampled. Higle and Sen [16] propose a cutting plane algorithm for two-stage stochastic linear programs with recourse. The method generates a statistical estimate of the objective value based on randomly generated observations of the random variables.

Generation of data trajectories or path-based methods

Principle Path-based methods start by generating complete paths or scenarios, by evaluating the stochastic process by means of econometric or time series models. The result is a set of scenarios, characterized by a fan-like structure in the scenario tree [3]. The scenarios can be clustered together in all-but-the-last period if required. Regression methods may be used to fit the model to available data. Examples include the well-known auto-regressive moving average models (ARMA, ARIMA), generalized auto-regressive conditional heteroskedasticity models (GARCH) and (Bayesian) vector auto-regressive models (VAR, BVAR).

Pros & cons These techniques are capable of rendering scenarios for statistical dependent random processes [3]. These methods generally require the assumption of a Gaussian stochastic variable, but can accommodate otherwise distributed stochastic variables through distribution transformations (e.g. Conejo [3], Sharma et al. [17], Pinson [13], Ma et al. [14]). This does however increase the complexity of the method.

Examples Models that belong to the family of autoregressive and moving average model techniques, such as the ARIMA (autoregressive integrated moving average model) described in Conejo et al. [3], are path-based methods. The ARMA model is applied in the context of wind power by Sharma et al. [17], with a distribution transformation from Gaussian to Weibull.

Property or moment matching

Principle These methods generate a discrete distribution (under the form of a set of scenarios) that satisfy a prefixed set of statistical properties (e.g. moments, correlation matrix, percentiles, . . .) of the original distribution. The structure of the scenario tree may be predefined. The best-known example of such a technique is the so-called *moment-matching method*, as described by Hoyland and Wallace [18]. Other methods, such as *pricipal component analysis*, are used in finance and econometric, but are to the best of our knowledge not used to generate scenarios for wind power forecast errors. Key in all the before mentioned methods are the underlying statistical approaches.

Pros & cons In a property matching scenario generation technique, no explicit knowledge of the uncertain variables pdf is needed. Only the (statistical) properties of the uncertain variable that are considered important by the modeler need to be known, making this technique especially useful when little information is available [2] Moreover, these properties can be 'matched' with a relatively low number of scenarios (in contrast to the large number of scenarios typically obtained in sampling techniques). However, it is very difficult to assess the importance of a statistical property in a stochastic programming problem ex-ante. In addition, the statistical properties of the uncertain variable might be met via multiple distinctly different sets of scenarios of the same cardinality. The modeler will than have to choose between these different sets. Hochreiter and Pflug [19] illustrate that the moment-matching technique may lead to strange results, as the difference between the moments of the distribution and the proposed approximation never lead to a real distance or metric between the two. As these methods generally do not guarantee convergence, increasing the number of scenarios to improve the stability or to reduce the bias of the method, is not guaranteed to help [7].

Advantages of property matching techniques are the possibility to include inter-period statistical dependencies and the possibility impose the inclusion of a worst-case scenario as a statistical property to satisfy. For the latter advantage, the same remark as before holds: it is next to impossible to ex-ante specify the worst-case wind power forecast error scenario.

Examples [82,83] in Conejo et al. [3], Hyland [18], Models based on regression analysis or principal component analysis.

Optimal scenario generation techniques based on probability metrics

Principle Given a specified structure for the required scenario tree, probability metrics such as the Wasserstein or Kantorovich distance are used to express the difference between two distinct scenarios. The scenarios are then generated in order to minimize the approximation error, i.e. to minimize the difference between the solution of the stochastic program considering the true distribution of the uncertain variable and the solution of the stochastic program considering a discrete representation of the same uncertain variable. As shown by Pflug [9], the probability distance between some scenarios is bounded by some integral of the norm of the difference between two scenarios, which allows to minimize this metric rather than solving the full stochastic program.

Pros & cons Although empirical evidence suggests that this type of scenario generation will lead to optimal scenario trees (i.e. with the smallest possible approximation error for that cardinality), a formal proof of convergence not available. The method however is complex. The structure of the tree is fixed, but can be optimized simultaneously by regarding the problem as a facility location problem, in which the nodes of the tree are the locations of the facilities that need to be optimized. However, this leads to extremely high calculation times. As will be illustrated below, the major drawback of this method with respect to its application for wind power forecast errors lies in the calculation of (the upper bound on) the Kantorovich distance based on a norm of the difference between two scenarios. As a result, all inter temporal information (i.e. the variability of a scenario) is lost. This could be solved by considering the true approximation error, but this requires solving the true stochastic programming problem.

Examples Pflug [9]

3.2 The proposed scenario generation technique

The aim of this WPFE scenario generation techique is to generate a relatively small set of scenarios that is a good representation of the statistical properties of the WPFE and captures all events that will significantly affect (the objective function of) the unit commitment problem. We propose a scenario generation technique based on Pinson et al. [13], refined by Ma et al. [14]. The proposed method combines path-based methods and conditional sampling techniques. We assume that the statistical analysis, as outlined in Bruninx et al. [11], has been performed on available wind power forecast error data. The result of this analysis is a description of the probability density function (pdf) and the cumulative probability density function (cdf) of the error for a number of predefined forecast power bins or intervals.

The proposed method then consists of **three main steps**. A graphical overview can be found in Fig. 2. First, a random number generator is used to **generate random realizations of a multivariate Gaussian distribution** $N(\mathbf{0}, \mathbf{\Sigma})$. The covariance matrix $\mathbf{\Sigma}$ represents the interdependence structure of the forecast errors over the planning horizon PH. Each element of the covariance matrix $\mathbf{\Sigma}$ is estimated using an exponential covariance function [14]:

$$\Sigma_{m,n} = exp\left(-\frac{|n-m|}{\delta\left(NS^{-1}\sum_{j=t+1}^{t+t^{PH}} g_j^F\right)\epsilon}\right) \tag{4}$$

with ϵ a range parameter controlling the strength of the correlation of WPFE at different lead-times, here set to 75, and δ a correction function ranging depending on the average wind power forecast g^F over the planning horizon $NS^{-1}\sum_{j=t+1}^{t+t^{PH}} g_j^F$. This correction function allows accounting for the higher variability for intermediate forecasts. Indeed, due to the non-linear nature of the power curve of a wind turbine or farm, forecast errors tend to be larger and more erratic at intermediate forecast levels. This correction function is optimized in order to minimize the approximation error of the resulting discrete probability density function and the original probability density function. At the end of the first step, we have obtained a set of equiprobable scenarios $\mathbf{Z}$ with a predefined cardinality. These scenarios are inter temporal consistent, i.e. their variability reflects the variability of the wind power forecast error, as will be illustrated in Section 5.

In the **second step**, the scenarios, generated from a Gaussian distribution, are **transformed to wind power forecast error scenarios** using the cdfs of the WPFE. From statistical theory, it is known that the inverse transformation of a cumulative probability distribution yields a uniform distribution on the interval $[0, 1]$ [13,14]. In addition, the cdf of the Gaussian distribution in step 1 is a uniform distribution on the interval

[0, 1]. Therefore, one can use the following transformation for each node (value $Z_{t,s}$) in each scenario s:

$$WPFE_{t,s} = F_t^{-1}\left(\Phi\left(Z_{t,s}\right)\right) \tag{5}$$

$$\Phi(Z_{t,s}) = \int_{-\infty}^{Z_{t,s}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \tag{6}$$

In these equations F_t^{-1} stands for the inverse cumulative probability density function of the forecast error at time step t, which is dependent on the forecasted wind power at that moment. $Phi(Z_{t,s})$ is the cdf of the Gaussian distribution (random variable $Z_{t,s}$). This transformation is performed on all values on time step t for all scenarios s with the cdf F_t, dependent on the forecast level. This process is repeated for all time steps t. For each time steps, the cdf corresponding to the forecast level is selected. This transformation is illustrated in Fig. 3.

Finally, the probability of each event in the scenarios is calculated via a Lvy alpha-stable distribution, fitted to the historical wind power forecast error data according to Bruninx et al. [11], in order to **remove improbable scenarios**. As the scenario generation technique cannot take into account all information on the variability of the error, some impossible transitions (e.g. too high variability) can occur in the generated scenarios. To avoid that the power system is exposed to these challenging, but impossible scenarios, these scenarios should be removed from the considered set. The probability of each event in each scenario $prob_{j,s}$ is calculated, over the simulated planning horizon PH, from

$$\forall s, \forall j: \quad prob_s = p_1(\epsilon_{j,s}|\epsilon_{j-1,s}) \cdot p_2(\epsilon_{j,s}|g^F_{j,s}) \tag{7}$$

where $p_1(\epsilon_{j,s}|\epsilon_{j-1,s})$ is the conditional pdf of the occurrence of an error $\epsilon_{j,s}$ given the previous error $\epsilon_{j-1,s}$. Similarly, $p_2(\epsilon_{j,s}|g^F_{j,s})$ is the conditional pdf of an error $\epsilon_{j,s}$ occurring given the wind power forecast $g^F_{j,s}$. The index s indicates the scenario, the index j indicates the time step. Scenarios that contain only events with a non-zero probability are retained. All other scenarios are removed. In this probability calculation, it is assumed that the wind power is perfectly known at the end of the previous time period ($\epsilon_{0,s} = 0$).

The major advantage of this process is that it is much easier to represent the joint multivariate normal distribution (which allows us to take into account inter temporal effects within a scenario) compared to representing e.g. a multivariate stable distribution directly. In addition, we take the variability of the WPFE correctly into account. Moreover, the computational complexity of the method remains acceptable, in contrast to e.g. optimal scenario generation techniques based on probability metrics.

Compared to the literature (and especially compared to Pinson [13] and Ma et al. [14]), we have improved the methodology on two fronts. First, we introduce a correction for the covariance matrix that captures the larger and more erratic WPFE at intermediate forecast errors. Second, an ex-post evaluation of the resulting scenarios is performed in order to detect improbable scenarios. Removing these scenarios avoids the overestimation of the impact of wind power forecast errors.

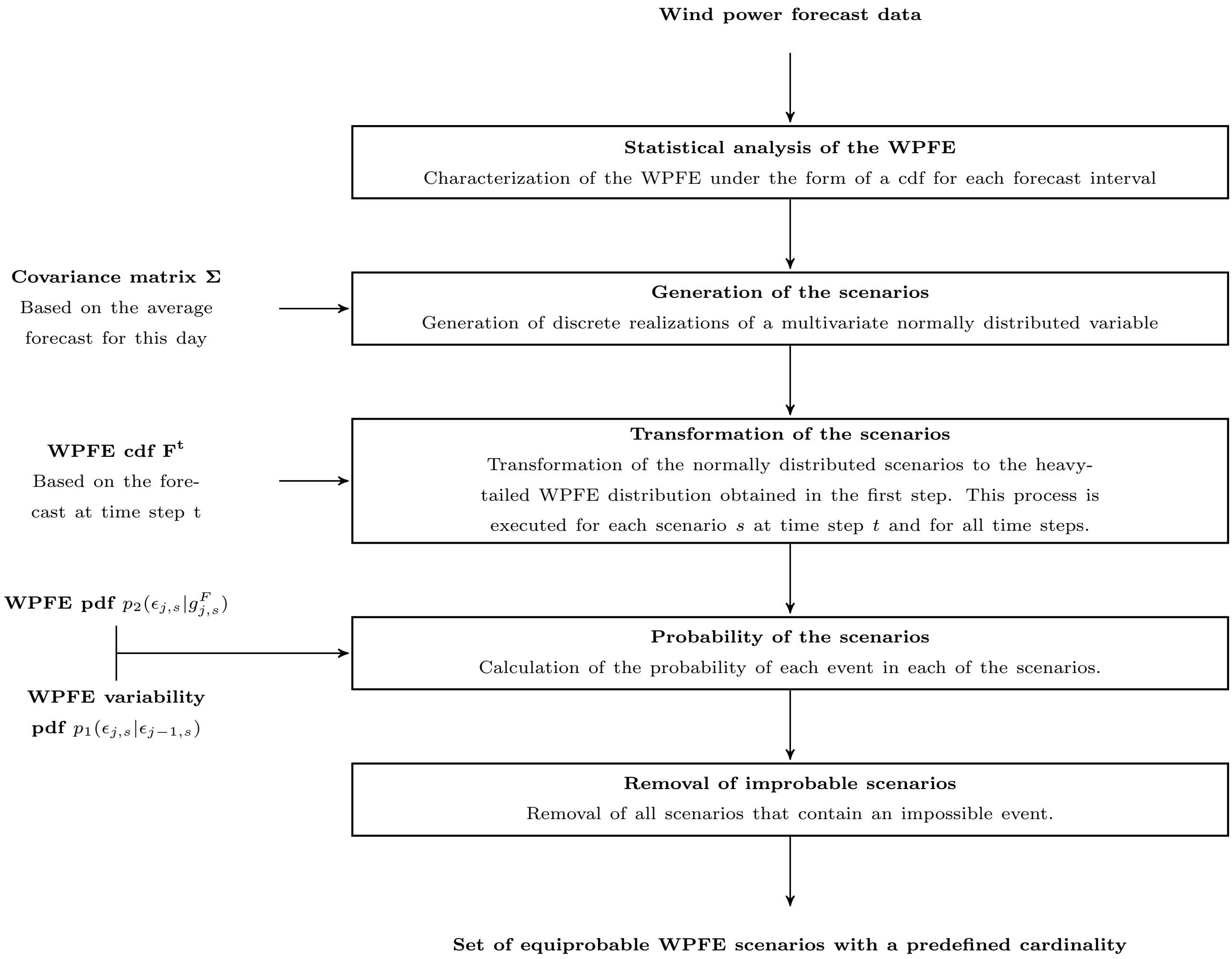


Figure 2: Visual representation of the proposed scenario generation method. The required input of at each step of the methodology is indicated on the left.

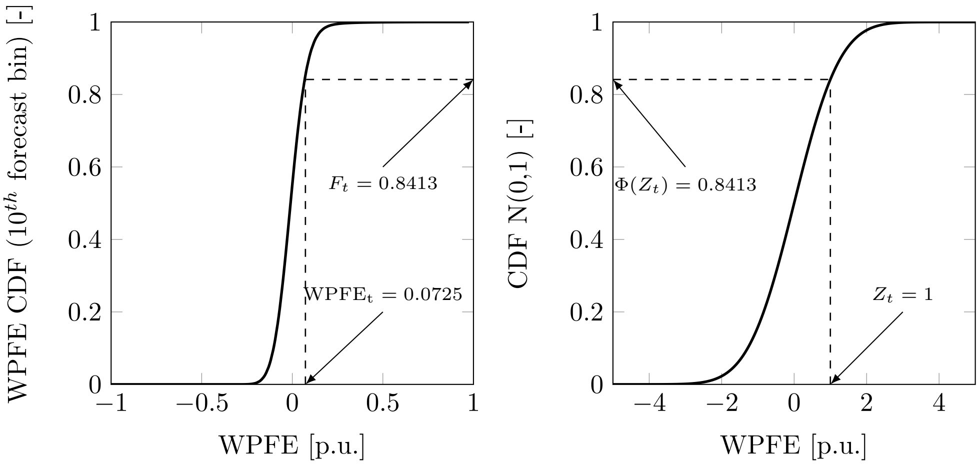


Figure 3: The generated realizations of the normally distributed variable Z_t is transformed via the cdf of the WPFE F_t, selected from the statistical analysis based on the wind power forecast at that time step t. This figure is based on Ma et al. [14].

4 Scenario tree reduction

In the following section, scenario tree reduction methods will be discussed. Again we will focus on those methods that have been or could be applied to wind power forecast errors. In our analysis, we will assume that a large set of scenarios is given, characterizing the statistical variable under consideration. Note that some authors combine scenario tree generation and reduction in a one-step approach [REF needed]. However, scenario generation and reduction methods typically different objectives. Scenario generation methods will seek to characterize the stochastic variable at hand via a set of discrete realizations of said variable. However, as stated above, considering this full set will typically render the stochastic optimization problem at hand intractable, or will yield high computation times. Regaining the tractability of the problem, without losing the stochastic information that is represented in the scenarios is the goal of the scenario reduction algorithm. One seeks to reduce the number of scenario that need to be considered in the optimization, without sacrificing solution quality: the difference in objective function of the optimization problem solved on the reduced set of scenario and on the full set of scenarios[2] should be as small as possible.

4.1 Overview of the scenario tree reduction methods

The following overview is based on Heitsch and Romisch [20] and extended with information from [2]. For an overview of the scenario reduction techniques prior to 2000, the reader is referred to the latter. For now, we will distinguish between the following scenario reduction techniques:

- Sequential importance sampling;
- Moment-matching based techniques;
- Optimal approximation methods based on probability metrics;
- Heuristics.

However, no widely accepted classification exists. Some scenario reduction methods do not fit in these(wide) categories and are not discussed below. Furthermore, note that scenario reduction techniques often result in optimization problems themselves. As these problems are often characterized by many local maxima and minima and are thus difficult to solve exactly. In many cases, heuristics algorithms have to be employed to approximate the solution. We will however classify these heuristics based on the original approach (thus optimization problem) they refer to.

Sequential importance sampling

Principle Given an initial scenario tree structure, a sequential importance sampling scenario generator will select some sample paths from this scenario tree. The stochastic program is solved and the nodal values of the importance sampling criterion are evaluated. Based on these values, a new sample tree structure is identified. As such, a reduced scenario tree and a solution of the stochastic program will be obtained iteratively. Various sampling rules can be defined. This technique is mostly employed for multi-stage stochastic problems.

Pros & cons The efficiency of the method depends heavily on the adopted sampling rule.

Examples A possible importance sampling criterion is the expected value of perfect information (EVPI)[3] – see [2,21,22]. After every iteration, the EVPI in each node of the scenario tree is updated. At termination, a complex tree structure is typically generated. Ideally, the optimal here-and-now solution becomes independent of possible additional scenarios sampled from the initial scenario tree. Dupacova [2] expands this technique to preserve the moments of the underlying distribution on each node and as such it becomes a moment-matching technique (see below).

Moment-matching principle

Principle The goal in this type of scenario-reduction technique is to match specified statistical properties of the reduced set of scenarios and those of the original (possibly infinite) set of scenarios. To this end, a measure of distance such as the squared norm between the the statistical properties of the constructed, reduced distribution and the original distribution. As this problem is often

[2]Note that this solution might be impossible to obtain.

[3]The expected value of perfect information or EVPI is defined as the difference between the optimal value of the stochastic program (here-and-now decision) and the associated expected value of the equivalent perfect foresight, deterministic, wait-and-see problem.

non-convex, heuristics have been developed to approximate the optimal solution [18]. Note that this method can also be used as a pure scenario generation technique, if one starts from continuous distributions instead of a discrete scenario tree.

Pros & cons Preservation of the statistical properties of the original distribution of the random parameters. The user can specify the statistical parameters that he or she finds important in the problem at hand. Hochreiter and Pflug [23] illustrate that the moment-matching technique may lead to strange results, as the difference between the moments of the distribution and the proposed approximation never lead to a real distance or metric between the two. The probability metrics are therefore further explored in [23]. Dupacova [2] states that moment-matching techniques are especially useful when little information is available on the underlying statistical process. As these methods generally do not guarantee convergence, increasing the number of scenarios to improve the stability or to reduce the bias of the method, is not guaranteed to help [7].

Examples [18], [24]: In Hoyland and Wallace [18], the reduction problem of multi-period scenario trees is split up in single period problems. This sequential simplifies the non-convex multi-period optimization to non-convex single-period optimizations, thus requiring less computational effort per period. However, it does not allow to control the statistical parameters of all outcomes.

Optimal approximation methods based on probability metrics

Principle These methods stems froom probability metrics such as the Fortet-Mourier and the Kantorovich distance[4] [4]. Especially the Kantorovich functional can be seen as a measure of the probabilistic distance between two scenarios and can thus be used to determine the optimal reduced set of scenarios to minimize the probabilistic difference between the original and the reduced set of scenarios (see [4]). This optimal scenario generation or reduction problem his however often an non-differentiable non-convex problem that is often too large in scale to be practical in many practical applications [4, 9]. Therefore this optimal set of scenarios is often approximated via a heuristic model [4].

Pros & cons Stability can be guaranteed if a certain family of Kantorovich or transportation metrics is selected. From Sharma [17] it seems that this can only be applied to two-stage stochastic programs.

Examples In Dupacova [4], two algorithms based on the Kantorovich functional are introduced: the backward and the forward reduction technique. The algorithms are further refined in Gröwe-Kuska et al. [25] and Heitsch and Römisch [20, 26]. Some modifications are made based on stability results and via the inclusion of a filtration distance in Heitsch and Römisch [27, 28]. The algorithms as described in [25] for huge scenario sets are included in the GAMS library SCENRED and used in the WILMAR project [29]. Note that Pflug also employs these probability metrics to construct optimal scenario trees [9, 23, 30].

4.2 The proposed scenario reduction method: an improved probability distance-based algorithm

The generated set of scenarios serves as the input for a modified *forward selection* scenario reduction technique [25]. Below we will not discuss the theoretical background of this algorithm in detail. This can be found in the literature, e.g. [2–4, 31]. The aim of this algorithm is to select a set of scenarios with a predefined, typically low cardinality, that will yield a stable solution of the stochastic unit commitment problem – i.e. reducing the number of scenarios with a minimal difference in objective value and optimal solution compared to the original problem (see Section 2). This 'selection' problem can be formulated in itself as an optimization problem, if one succeeds in defining some metric that describes the difference between two sets of scenarios. In that case, scenario reduction boils down to an optimization problem in which one tries to minimize this so-called *probability distance metric* between the original set of scenarios and a new, reduced set of scenarios with a predefined cardinality.

In this line of thought, researchers have considered the *Monge-Kantorovich mass transport problem* as a description of the scenario reduction problem. Without going into details, one can think of this Monge-Kantorovich mass transport problem as an optimization problem that defines the optimal, in this case minimal, probability mass transportation that needs to occur to map one probability density function on to another [32]. In the context of two (discrete) probability density functions formed by two sets of scenarios, the solution to this

[4] Depending on the precise definition, alternative notations such as the Kantorovich-Rubinstein or Wasserstein distance are found in literature.

optimization problem represents the optimal redistribution of probability over a reduced set of scenarios that minimizes the statistical distance between the original set. As shown in [4], the Monge-Kantorovich mass transport problem can be formulated as follows for two sets of discrete scenarios Ω_s (elements ω') and Ω (elements ω) with finite probability distributions Q and Q' [3, 31]:

$$D_K(Q, Q') = \min\{\sum_{\omega,\ \omega'} c(\omega, \omega')\eta(\omega, \omega')\} \tag{8}$$

$$\text{subject to} \quad \forall \omega, \omega' \quad \eta(\omega, \omega') \geq 0 \tag{9}$$

$$\forall \omega' \quad \sum_{\omega} \eta(\omega, \omega') = \pi_{\omega'} \tag{10}$$

$$\forall \omega \quad \sum_{\omega'} \eta(\omega, \omega') = \pi_{\omega} \tag{11}$$

In this formulation $c(\omega, \omega')$ is referred to as the cost function. In the case of a pure Kantorovich distance D_K this is a norm. π_ω and $\pi_{\omega'}$ represent the probability of scenario ω, ω' in sets Ω and Ω_s respectively. $\eta(\omega, \omega')$ relates to the joint probability distributions defined on $\Omega \times \Omega$. The Kantorovich distance can be seen as a measure of the probabilistic distance between two scenarios and can thus be used to determine the optimal reduced set of scenarios to minimize the probabilistic difference between the original and the reduced set of scenarios.

The Kantorovich distance can, for two stage problems and under mild assumptions[5] [3, 31], be simplified to

$$D_K(Q, Q') = \sum_{\omega \in \Omega \setminus \Omega_s} \pi_\omega \min_{\omega'} c(\omega, \omega') \tag{12}$$

Eq. (12) often results in a non-differentiable non-convex problem that is too large in scale to be practical in many applications [25]. Researchers have developed several heuristics to solve this problem in reasonable time, of which the *fast forward* and *backward algorithm* developed by Dupačová et al. [2, 4] is the best-known. In this paper, we will, similarly as in [31], focus on the *fast forward algorithm*[6].

The original iterative, greedy *fast forward* algorithm will select a set of scenarios Ω_s with a predefined cardinality from an original set of scenarios Ω in order to minimize the Kantorovich distance K_D between the probability density distributions Q' and Q of the reduced and original set of scenarios respectively [3, 4, 31]. Starting from a empty set, scenarios are selected one-by-one until the predefined number of scenarios is reached or, in some cases, a certain Kantorovich distance is reached. A detailed description of the algorithm can be found in [4]. The cost function $c(\omega, \omega')$ becomes a norm in the original scenario reduction algorithm:

$$c(\omega, \omega') = \|WPFE_\omega - WPFE_{\omega'}\| \tag{13}$$

However, as this scenario reduction algorithm only looks at the amplitude of the error to differentiate between the different scenarios. Other characteristics of the error, such as the variability, are not taken into account. As will be shown in the box *Intermezzo: a methodological illustration* below, the original scenario reduction method cannot differentiate between variable and constant scenarios. However, these elements might have a significant impact on the objective function and first-stage decisions that need to be taken. For example, in the case of variability, the possible absence of highly variable scenarios might lead to a unit commitment schedule that is not able to facilitate strongly fluctuating injections of wind power due the ramping constraints of the scheduled units. Increased curtailment or load shedding and the associated increase in operational costs will be the result.

Therefore, we here propose a new cost function $c(\omega, \omega')$, similar to the one proposed in [3, 31]. The goal (in both cases) is to capture the effect of the addition of a certain scenario to the reduced set of scenario on the resulting objective function. To capture this effect, the proposed cost function is $c(\omega, \omega')$ is given by

$$c(\omega, \omega') = |z_\omega - z_{\omega'}| \tag{14}$$

in which z_ω represent the objective value of the equivalent deterministic, one scenario equivalent of the stochastic unit commitment problem at hand, in which the set of wind power forecast error is replace by the realization ω. Note that in this paper, in contrast to the proposed cost function proposed by Morales et al. [3, 31] the first stage decision variables are not fixed in the one-scenario equivalent problem. Morales et al. employ the

[5] I.e., the stochasticity should be pertained to right-hand sides of the stochastic problem at hand. This assumption holds for wind power forecast errors.

[6] Note that, as pointed out by Morales et al. [31], these heuristics do not guarantee to yield to result in the set of scenarios with the lowest possible Kantorovich distance to the original set of scenarios. Moreover, this reduced set might lead to a solution of the optimization problem with an objective value far from the objective value of the optimization problem considering the full set of scenarios, or a set of scenario that yields a stable solution (see Section 2). However, the literature suggest that the solutions obtained on reduced sets of scenarios, obtained via the *fast forward algorithm*, yield satisfactory results.

same cost function, but calculate the cost of the one-scenario equivalent problem with fixed first-stage decision variables. The value of these first-stage decision variables have been obtained form the expected value problem corresponding to the stochastic unit commitment problem, a deterministic unit commitment problem in which the wind power forecast error scenarios are replaced by the expected value of the forecast error. However, for the stochastic unit commitment problem we propose cost function Eq. (14) because this approach allows us to capture the possible benefits of different decisions on the first stage variables, while the proposed cost function of Morales et al. [31] fixes the first stage variables. The effect of the difference in approaches will be discussed in detail in the methodological example below and in Section 5.

With the objective function (Eq. (12)) and the cost function definition (Eq. (14)) in mind, we can now use the *fast forward algorithm* [4] to determine the elements (scenarios) in set Ω_s that minimize the Kantorivich distance between the original set and the reduced set of scenarios. This algorithm, developed by developed by Dupačová et al. [4,25] and illustrated in Fig. 4. In the first step, the cost function $c(\omega, \omega')$ is calculated for each pair of scenarios. In the second step, one needs to select the first scenario that will form the basis of the reduced set of scenarios. In the *fast forward algorithm* the scenario that is most equidistant from the other scenarios in the set is chosen as the first scenario in the reduced set. This scenario can be obtained by solving

$$\omega_1 = \arg\{\min_{\omega' \in \Omega} \sum_{\omega \in \Omega} \pi_\omega c(\omega, \omega')\} \tag{15}$$

As of the third step, one will start iterating, adding a scenario in each iteration until the defined cardinality of the reduced scenario set N_s is reached. The selection of these scenarios is based on the following equation for each step i:

$$\omega_i = \arg\{\min_{\omega' \in \Omega \setminus \Omega_s^i} \sum_{\omega \in \Omega \setminus \Omega_s^i \setminus \omega'} \pi_\omega \min_{\omega'' \in \Omega_S^{i-1} \bigcup \omega} c(\omega, \omega'')\} \tag{16}$$

in which Ω_S^i is the set of scenarios that are selected before iteration i of the algorithm. After repeating this step in the algorithm $N_s - 1$ times, the reduced set of scenarios Ω_s contains N_s scenarios. As a final step, the probability of not selected scenarios (set $\Omega \setminus \Omega_s$) is optimally redistributed over the retained scenarios (set Ω_s):

$$\pi_\omega = \pi_\omega + \sum_{\omega' \in J(\omega)} \pi_{\omega'} \tag{17}$$

with $J(\omega)$ the set of scenarios $\omega' \in \Omega \setminus \Omega_s$ such that $\omega = \text{argmin}_{\omega' \in \Omega_s} c(\omega'', \omega)$. This means that the probability of a scenario that is not selected is added to the probability of the scenario that is most alike according to the cost function $c(\omega, \omega')$.

As shown in the methodological illustration below and Section 5, the objective value of the deterministic equivalents is the best available proxy for the impact of the wind power forecast error scenarios on the objective function of the SUCM and thus the only relevant measure to include or exclude a scenario in the stochastic optimization. As such, we succeed in selecting a balanced reduced set of scenarios. For a detailed discussion on the technique and its added value, see Section 5 and [1].

Set wind power forecast error scenarios Ω

$\downarrow$

Step 1

Calculate cost function $c(\omega, \omega')$ for each scenario pair

$$\forall \omega, \omega' : \quad c(\omega, \omega') = |z_\omega - z_{\omega'}|$$

$\downarrow$

Step 2

Select the first scenario ω_1

The first scenario is the most equidistant scenario from all other scenarios in the set Ω:

$$\omega_1 = \arg\{\min_{\omega' \in \Omega} \sum_{\omega \in \Omega} \pi_\omega c(\omega, \omega')\}$$

$\downarrow$

Step 3

Repeat selection $N_s - 1$ times until set Ω_s reaches cardinality N_s

Select scenarios based on the Kantorovich distance between the scenarios that are already selected Ω_S^{i-1} and the remaining scenarios in the original set Ω:

$$\omega_i = \arg\{\min_{\omega' \in \Omega \setminus \Omega_s^i} \sum_{\omega \in \Omega \setminus \Omega_s^i \setminus \omega'} \pi_\omega \min_{\omega'' \in \Omega_S^{i-1} \cup \omega} c(\omega, \omega'')\}$$

$\downarrow$

Step 4

Redistribute the probability of the scenarios

Redistribute the probability of not selected scenarios over the selected scenarios that are most alike according to the cost function $c(\omega, \omega')$:

$$\pi_\omega = \pi_\omega + \sum_{\omega' \in J(\omega)} \pi_{\omega'}$$

with $J(\omega)$ the set of scenarios $\omega' \in \Omega \setminus \Omega_s$ such that $\omega = \text{argmin}_{\omega' \in \Omega_s} c(\omega'', \omega)$.

$\downarrow$

Reduced set of WPFE scenarios Ω_s with a predefined cardinality N_s

Figure 4: Illustration of the *fast forward scenario reduction algorithm* as developed by Dupačová et al. [4, 25].

Intermezzo: a methodological illustration

Scenario reduction cannot be decoupled from the stochastic model in which one will use the scenarios. Therefore, we will consider a methodological, stylized version of a stochastic unit commitment problem as described below. In Section 5 we will show that these conclusions also hold for the full stochastic unit commitment problem in power systems of a realistic size.

A stylized stochastic unit commitment problem Consider a power system with two generators, A and B, characterized by their start-up costs SC (SC_A = 50,000 EUR, SC_B = 50,000 EUR), direct fuel costs FC (FC_A = 50 EUR/MWh, FC_B = 60 EUR/MWh) and ramping costs RC related to changes in output (RC_A = 10 EUR/MW, RC_B = 0 EUR/MW). The optimization period is four hours long (index i), which leads to the following objective function:

$$\min \quad SC_A \cdot z_A + SC_B \cdot z_B + \sum_{\omega} \pi_\omega \left[\sum_{i=1}^{4} (FC_A \cdot g^A_{\omega,i} + FC_B \cdot g^B_{\omega,i}) + \sum_{i=2}^{4} (RC_A \cdot |\Delta^{g^A_{\omega,i}}_{g^A_{\omega,i-1}}| + RC_B \cdot |\Delta^{g^B_{\omega,i}}_{g^B_{\omega,i-1}}| \right] \tag{18}$$

In this equation, index ω (set Ω) indicates the scenarios, each with probability π_ω. $g^A_{\omega,i}$ is the output of power plant A on hour i under scenario ω. z is a binary variable indicating the on/off status of the power plant, independent of the scenario ω. This optimization is subjected to two constraints. First, the demand for electric power, which is always equal to 1,000 MW, must be met in each hour by the output of the generators and some uncertain wind power:

$$\forall i, \forall \omega \quad g_{\omega,i} + g_{\omega,i} + WIND_{\omega,i} = 1,000 \;\; MW \tag{19}$$

Second, a unit can only produce power if it is switched on at the beginning of the period:

$$\forall i, \forall \omega \quad 0 \leq g_{\omega,i} \leq 1,000 \;\; MW \cdot z_A \qquad\qquad \forall i, \forall \omega \quad 0 \leq g_{\omega,i} \leq 1,000 \;\; MW \cdot z_B \tag{20}$$

Furthermore, consider 4 scenarios, as shown in Fig. 5. Scenario 1 corresponds to a stable output of 750 MW, scenario 2 to a stable output of 250 MW. In scenario 3, the output varies between 0 and 1,000 MW. Scenario 4 is the reverse of scenario 3. Each of these scenarios occurs with a probability of 1/4.

The optimal solution to the stylized stochastic unit commitment problem The solution of the stochastic optimization problem described above can be determined at sight, using Table 1 below. The expected operational cost, i.e. the objective value of the proposed stochastic unit commitment problem would be 187,500 EUR if generator A would be switched on and 170,000 EUR if generator B would be switched on. Switching both generators would reduce fuel costs and eliminate ramping costs, but the increased start-up costs would render this solution more expensive. The optimal solution for the first stage variables thus equals $z_A = 0$ and $z_B = 1$, with an expected cost of 170,000 EUR.

Scenario reduction Imagine we would reduce the number of scenarios considered in this optimization. Which scenarios would we select with the *fast forward algorithm*? As shown above, the cost function will determine which scenario will be selected, as it will determine the Kantorovich distance between the reduced and original set of scenarios. In the **first step of the scenario reduction algorithm**, we calculate the cost function $c(\omega, \omega')$. We distinguish $c_D(\omega, \omega')$, the cost function as proposed by Dupačová et al. [4,25]; $c_M(\omega, \omega')$, the cost function as proposed by Morales et al. [31] and $c_B(\omega, \omega')$, the cost function proposed in this paper:

$$c_D(\omega, \omega') = \begin{pmatrix} 0 & 2,000 & 2,000 & 2,000 \\ 2,000 & 0 & 2,000 & 2,000 \\ 2,000 & 2,000 & 0 & 4,000 \\ 2,000 & 2,000 & 4,000 & 0 \end{pmatrix} \quad \rightarrow \quad d_D(\omega) = c_d(\omega, \omega') \cdot \begin{pmatrix} 0.25 \\ 0.25 \\ 0.25 \\ 0.25 \end{pmatrix} = \begin{pmatrix} 1,500 \\ 1,500 \\ 2,000 \\ 2,000 \end{pmatrix}$$

$$c_M(\omega, \omega') = \begin{pmatrix} 0 & 100,000 & 125,000 & 125,000 \\ 100,000 & 0 & 25,000 & 25,000 \\ 125,000 & 25,000 & 0 & 0 \\ 125,000 & 25,000 & 0 & 0 \end{pmatrix} \quad \rightarrow \quad d_M(\omega) = c_M(\omega, \omega') \cdot \begin{pmatrix} 0.25 \\ 0.25 \\ 0.25 \\ 0.25 \end{pmatrix} = \begin{pmatrix} 87,500 \\ 37,500 \\ 37,500 \\ 37,500 \end{pmatrix}$$

$$c_B(\omega, \omega') = \begin{pmatrix} 0 & 100,000 & 70,000 & 70,000 \\ 100,000 & 0 & 30,000 & 30,000 \\ 70,000 & 30,000 & 0 & 0 \\ 70,000 & 30,000 & 0 & 0 \end{pmatrix} \quad \rightarrow \quad d_B(\omega) = c_B(\omega, \omega') \cdot \begin{pmatrix} 0.25 \\ 0.25 \\ 0.25 \\ 0.25 \end{pmatrix} = \begin{pmatrix} 60,000 \\ 40,000 \\ 25,000 \\ 25,000 \end{pmatrix}$$

The cost function $c_D(\omega, \omega')$ is easily calculated as the sum of the absolute difference between the wind power scenarios. For example, $c_D(1, 2)$ equals $4 \cdot (750 \text{ MW} - 250 \text{ MW}) = 2,000$ MW. Note that according to this cost function scenario 1 and 2 are equidistant from scenarios 3 and 4. There is no difference between scenario 1 and 2, nor between scenario 3 and 4. Moreover, if one would remove scenario 4, one would not see any difference between the scenarios in terms of the Kantorovich distance $c_D(\omega, \omega')$. However, from our analysis above it is clear that scenario 3 and 4 are clearly distinct from scenario 1 and 2, as only those trigger the optimal first stage decision $z_B = 1$. In other words, this cost function does not allow to capture any variability and its impact on the objective functions. Cost functions $c_M(\omega, \omega')$ and $c_B(\omega, \omega')$ do allow to capture this effect, as will be illustrated below. Calculating $c_M(\omega, \omega')$ requires knowledge of the first stage solution, i.e. the value of z_A and z_B, in the solution of the one-scenario equivalent of the stochastic unit commitment problem *in which the wind power scenarios are replaced by their expected value.* One can easily verify that this expected value corresponds to a stable wind power output of 500 MW, which would yield a solution $z_A = 1$ and $z_B = 0$, yielding an operational cost of 150,000 EUR. $c_M(\omega, \omega')$ can then be calculated via Table 1 as the absolute difference between operational costs corresponding to $z_A = 1$ for each scenario. E.g. $c_M(2, 3)$ equals $|200,000 \text{ EUR} - 225,000 \text{ EUR}| = 25,000$ EUR. The cost function $c_B(\omega, \omega')$ can be derived from Table 1, as it is calculated as the absolute difference between the objective value of the one-scenario equivalent of the stochastic unit commitment problem. The wind power scenarios are retained. In this case, this means one should select the best solution for each scenario. E.g. $c_B(2, 3)$ equals $|200,000 \text{ EUR} - 170,000 \text{ EUR}| = 30,000$ EUR.

Based on these cost functions, one can calculate the Kantorovich distance d between the reduced set Ω_s for contain one of the scenarios and the remaining set. E.g., $d_B(1)$ should be interpreted as the Kantorovich distance between Ω_s, if it would contain scenario 1, and set $\Omega \setminus \Omega_s = \{2, 3, 4\}$. This Kantorovich distance now allows us to select the first scenario. For the cost function as proposed by Dupačová et al. [4, 25] the resulting Kantorovich distance does not allow for any differentiation between the scenarios 1 and 2 or 3 and 4: the algorithm does not show a preference in selecting between these scenarios. Moreover, this approach would force you to (arbitrarily) select scenario 1 or 2. Both of these scenarios yield the suboptimal soluton $z_A = 1$. Morales et al. [31] propose an alternative cost function, which yield a minimum Kantorovich distance if one selects scenario 2, 3 or 4. Although we have a good chance of selecting the correct scenario, the algorithm does not succeed in recognizing the high impact of scenarios 3 and 4 on the solution of the stochastic program. The cost function proposed in the present paper result in the selection of scenario 3 or 4: $\Omega_s^B = \{3\}$. Note that in both cases the Kantorovich distances are significantly different for each of the scenarios (except 3 and 4), but express different preferences towards the selection of each of the scenarios. The question now remains how we should judge the quality these results. One can do this by solving the stochastic unit commitment model considering the scenarios in Ω_s. Ideally, one would like to obtain the same solution from this optimization as one would if one solves the the stochastic unit commitment model all scenarios. This is the case for $\Omega_s^B = \{3\}$, while $\Omega_s^M = \{2\}$ would yield $z_A = 1$ and $z_B = 0$. The reason for this deviating result lies in the way the cost function is calculated. Morales et al. [31] base the cost used to characterize each scenario on the evaluation of the resulting deterministic equivalent employing a fixed first-stage decision. Although this allows to capture costs that depend on variability, such as the ramping costs in this simple problem, and thus to differentiate between scenarios with the same average value, one neglects the impact alternative first stage decisions might have. In this case, scenario 3 yields the highest cost due to the ramping costs if z_A is fixed to 1, making scenario 2 the 'average' scenario that will be selected. However, if one would allow z_A and z_B to take on other values, as we did to calculate $c_B(\omega, \omega')$, one sees the opportunity to reduce costs in scenario 3 or scenario 4 by setting z_B to 1. Scenario 3 (or 4) now becomes the most average scenario and will be selected in the first step in the algorithm presented in this paper.

In the **second step of the scenario reduction algorithm**, one needs to update the cost matrix $c(\omega, \omega')$. We will illustrate this for $c_B(\omega, \omega')$, but the reasoning is the same for the other cost functions. Using Eq. (16), the values of all rows except the row corresponding to the selected scenarios are updated. Each value in the cost matrix is compared to the value of the corresponding c_B value of the selected scenario. The minimum of these two values is retained:

$$c_B(\omega, \omega') = \begin{pmatrix} 0 & 70,000 & 70,000 & 70,000 \\ 30,000 & 0 & 30,000 & 30,000 \\ 70,000 & 30,000 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{21}$$

E.g. the new value $c_B(1, 2)$ is obtained as the $\min(c_B(1, 2), c_B(1, 3)) = \min(100000, 70000) = 70,000$. Note that the third row, corresponding to the selected third scenario, is unchanged. With this updated cost matrix the Kantorovich distance between set $\Omega_s = \{3\}$ and set $\Omega \setminus \Omega_s$ can be calculated as follows:

$$d_B(\omega \in \Omega) = \pi_{\omega \in \Omega \setminus \{w\}} \cdot c_B(\omega, \omega)^T$$

Applying this equation to this case:

$$d_B(\omega \in \Omega) = \begin{pmatrix} d_1 \\ d_2 \\ d_4 \end{pmatrix} = \begin{pmatrix} \pi_2 \cdot c_B(2, 1) + \pi_4 c_B(4, 1) \\ \pi_1 \cdot c_B(1, 2) + \pi_4 \cdot c_B(4, 2) \\ \pi_1 \cdot c_B(1, 4) + \pi_2 \cdot c_B(2, 4) \end{pmatrix} = \begin{pmatrix} 7,500 \\ 17,500 \\ 7,500 \end{pmatrix} \tag{22}$$

Hence, in the second step of this algorithm, we would select scenario 1 or 4: $\Omega_s = \{3, 1\}$.

In the **final step of the scenario reduction algorithm**, the probabilities are redistributed optimally based on the original cost matrix $c_B(\omega, \omega')$. Scenario 2 and 4 are closest to scenario 3, thus the probability of scenario 2 is added to scenario 3: $\pi_3 = 0.75$.

Relevance for 'real' unit commitment problems Unit commitment problems are riddled with costs that depend on more than just the amplitude of an stochastic variable such as the wind power forecast error. For example, the variability of a forecast error may lead to loss of load (with a very high cost) when insufficient flexible capacity had been committed by the unit commitment model. Scenario reduction techniques based on the absolute difference between the errors scenarios (Dupacova et al.), or based on the cost of those scenarios obtained from a dispatch considering a fixed expected-value unit commitment solution (Morales et al.), fail to recognize the importance of these scenarios in reaching the optimal unit commitment decision.

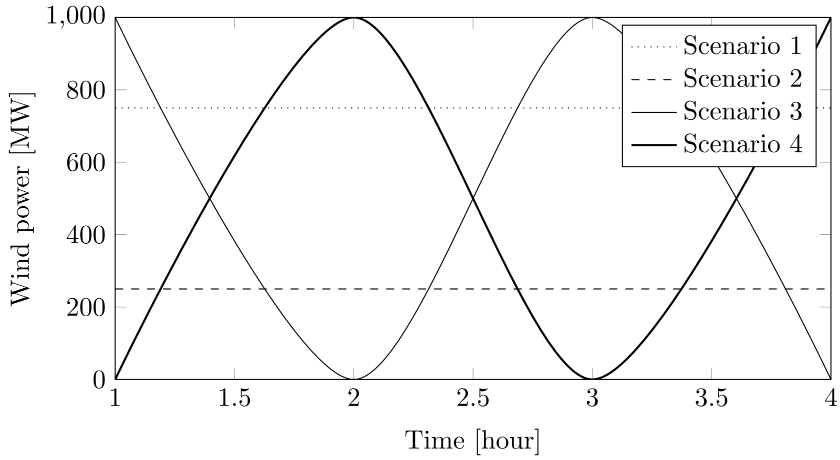


Figure 5: Four wind power scenarios are considered in this example. Scenario 1 corresponds to a stable output of 750 MW, scenario 2 to a stable output of 250 MW, scenario 3 and scenario 4 vary between 0 and 1000 MW.

hour	Scenario 1		Scenario 2		Scenario 3		Scenario 4	
	$z_A = 1$	$z_B = 1$	$z_A = 1$	$z_B = 1$	$z_A = 1$	$z_B = 1$	$z_A = 1$	$z_B = 1$
1	12,500	15,000	37,500	45,000	0	0	50,000	60,000
2	12,500	15,000	37,500	45,000	75,000	60,000	25,000	0
3	12,500	15,000	37,500	45,000	25,000	0	75,000	60,000
4	12,500	15,000	37,500	45,000	75,000	60,000	25,000	0
Total cost [EUR]	50,000	60,000	150,000	180,000	175,000	120,000	175,000	120,000
Start-up costs [EUR]	50,000	50,000	50,000	50,000	50,000	50,000	50,000	50,000

Table 1: Solution to the stochastic unit commitment problem of the methodological illustration. Each row corresponds to the operational cost in a specific hour (fuel costs + ramping costs) under a specific scenario. The total cost excludes the start-up costs. The situation in which the two power plants are simultaneously only has not been considered, as start-up costs would dominate the solution and yield a more expensive, hence sub-optimal solution.

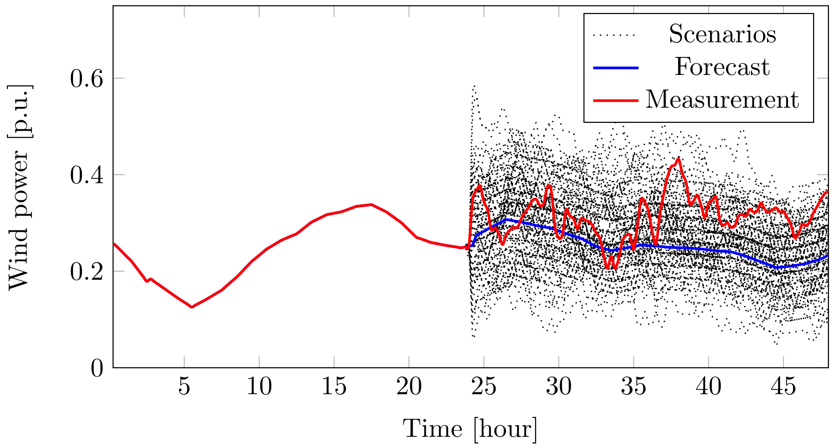


Figure 6: Throughout the discussion, the day with the residual demand closest to the yearly average will be studied. The figure above illustrates the forecasted and measured wind power production on that day (hour 24 and onward), as well as 100 WPFE scenarios generated for this wind power foreast.

5 Results & Discussion

In this section, the performance of the proposed scenario generation and reduction method will be investigated. The focus of the discussion will be on the criteria for an adequate scenario generation and reduction technique, as outlined above. As a case study, the Belgian power system, for which recent wind power forecast and production data is available from the Belgian TSO Elia NV [33], is proposed. The stochastic problem studied is a stochastic unit commitment problem or SUCM. In all SUCMs, one tries to find a schedule for a set of power plants, common to all scenarios, as such that a feasible dispatch is possible for all possible scenarios of an uncertain variable, in this case the WPFE. A feasible dispatch here means that the demand for electricity in each time step is met in all of the considered WPFE scenarios, while respecting all techno-economic constraints of the power plants. A full description of the stochastic program, the data and assumptions can be found online [34].

First, the scenario generation technique is shown to yield a very good representation of the wind power forecast error in the Belgian power system. We will generate scenarios from a fitted Lévy α-stable distribution, as discussed in [34]. The distribution of the errors is then compared to this distribution and the historical distribution of the variability of the error. The correspondence between the distribution of the WPFE data and the generated scenarios is found to be very good. This conclusion is confirmed via an event-based evaluation framework, as described by Pinson et al. [5].

Second, the effect of the removal of improbable scenarios – i.e. those that contain events with a probability equal to zero, calculated from the fitted distributions (see above) – on the correspondence of the scenarios with the data is shown to be limited. In the retained set of scenarios, the probability of some extreme events is somewhat underestimated.

Last, the importance of the scenario reduction method with respect to the stability of the stochastic program is illustrated. As will be shown in Section 5.3, the original fast forward scenario reduction technique does not yield a stable solution of the program, while the proposed scenario reduction technique does. For the specific day studied in this paper, 15 to 20% of the scenarios is sufficient to obtain a first and second stage objective value within a 5% range of the stable solution of the stochastic program. Computational efforts are however drastically reduced.

5.1 Scenario generation & the distribution of the wind power forecast errors

Starting from a forecasted wind power profile, the proposed scenario generation technique allows us to generate a certain number of scenarios for the error on this forecast. In Fig. 6, this is illustrated for the day with the residual demand, calculated as the demand for electricity minus the expected RES production, closest to the yearly average. In this figure, 100 scenarios are shown, as well as the forecast and the measured wind power. Although the scenarios may seem extreme, the measured wind power production shows that this kind variability seems to occur in reality as well. However, the question remains: how well do these scenarios represent the stochastic character of the WPFE? Below we will try to quantify how well these scenarios capture the wind power forecast error. In order to do so, we have generate 100 scenarios for each day in 2012, using historical wind power forecast data from the Belgian TSO Elia [33]. From these scenarios, we calculated the empirical probability density functions (pdf) and cumulative probability density functions (cdf) for each forecast power bin and for each 'variability' power bin. As described above and in [11], the WPFE data is sorted into different subsets according to the forecast at which they occurred. If then calculate the empirical pdf for each of these subsets, we can compare these the fitted, stable pdfs we obtained in [11]. Likewise, the variability can be studied by investigating the probability of an error of a certain amplitude, given the previous error. The data is again sorted in subsets, according to the previous error and the empirical distributions are calculated. These empirical distributions can then be compared to distributions obtained from historical data.

The distribution of the error for each forecast bin As the scenario generation technique employed in this paper starts from the distribution of the error given the forecast (bin), a perfect match between the original distribution and the distribution of the simulated scenarios should be obtained. The only precondition needed to obtain this match is that a modeler should generate sufficient scenarios. In this case, we have generated 100 scenarios. As illustrated in Fig. 7 for the 10^{th} forecast power bin – containing all errors observed for forecasts between 0.225 and 0.25 p.u. – the distribution of the generated error and the stable distribution (the input for the scenario generation method) is perfect. If the distributions of the forecast error over the various forecast power bins are merged (for details, see [**?**]), the distributions displayed in Fig. 8 are obtained. The probability density function and the cumulative distribution of the generated scenarios and stable distributions are identical, as expected. This is further illustrated in the so-called pp- and qq-plots in Fig. 8. While the former shows the cumulative distribution function of the stable distribution ('theoretical cdf') versus the empirical cumulative probability function ('ECDF') of the scenarios, the latter shows the percentiles of the stable distribution versus the percentiles calculated from the generated scenarios. In both cases, all points lie on a straight line through the points $(10^{-3}, 10^{-3})$) and $(1, 1)$, respectively $(-0.25, -0.25)$ and $(0.25, 0.25)$, indicating a perfect agreement between both data sets.

The distribution of the variability of the error In contrast to the distribution of the error in each forecast power bin, the correspondence of the distribution of the variability of the error is not guaranteed to correspond to the data we started from. Indeed, the only temporal relation that exist between errors is described by the covariance matrix. As described above, this is a mere estimate of the true covariance matrix, which will not guarantees an adequate representation of the error. In this paper, we will discuss the distribution of the error on time step t, given the error on the previous time step $t-1$. Similarly to the forecast power bins [11], we have defined 'previous error power bins' with a width of 0.025 p.u., which allows us to study this distribution in detail. For example, the 10^{th} power bin contains the distribution of all errors observed after an error between 0.25 and 0.275 p.u. on the previous time step. In Fig. 9 and 10 we examine the correspondence between the observed distribution ('Empirical') of the variability of the error and the variability of the error in the generated scenarios. In Fig. 9, the distribution of the errors in the 35th power bin (containing errors observed after an error between -0.15 p.u. and -0.125 p.u.) is shown. Although the agreement is not as good as for the errors in each forecast power bin, it is sufficient to state that this set of scenarios represents this distribution. Note that after a negative error, it is much more likely that this error persists than that this error will disappear or turn positive (Fig. 9). In Fig. 10, the overall distribution[7] of the variability of the generated scenarios is shown. As is evident from Fig. 10, the variability of the generated scenarios mimics observed variability adequately.

Event-based verification Up to this point, we have only focused on the distribution of the error, given the forecast or the previous error. However, some events, such as positive or negative forecast errors that persist over multiple time periods or prolonged negative or positive ramps in the forecast error, may be challenging for the power system that we are studying. Therefore, if these events occur in real-life, they should be reflected as well in the generated scenarios. To evaluate how well a set of scenarios captures the probability of a certain event, Pinson et Girard developed a event-based verification framework, based on so-called Brier-scores [5]. In their paper, Pinson and Girard discuss the need for and the added value of such an event-based scenario-evaluation approach in detail. They illustrate that while traditional probabilistic verification frameworks (such as the one presented above) have their merits, they should be complemented with event-based approaches to check if the scenario generation technique is capable of mimicking specific characteristics of the stochastic process.

In this paper, we have defined two types of events for which we will check the probability of their occurrence, which we will characterize via two functionals g_1 and g_2. The first type of event is the occurrence of a wind power forecast error of a certain magnitude ϵ that persists over a certain time period h. The functional g_1 describing this event can mathematically be expressed as

$$g_1(s_j, k, h, \epsilon) = \prod_{i=k}^{i=k+h} \mathbf{1}\{s_j^{t+i} \geq \epsilon\}, \tag{23}$$

[7]The distributions in the various 'previous error power bins' are merged as a probability-weighted sum, where the weights indicate the probability that an error belongs to such an error power bin. For details, see [**?**].

in which $\mathbf{1}\{\}$ is an indicator variable, which equals one if the expression between brackets is true and zero otherwise. s_j is the wind power forecast error scenario to which this functional is applied, while k is an auxiliary variable tho select the moment of interest in this scenario. The second event of interest relates to the variability of the forecast error. We will check the occurrence of a significant gradient in the wind power forecast error ϵ over a certain period of time h, which is expressed by functional g_2:

$$g_2(s_j, k, h, \epsilon) = \mathbf{1}\{\max_{i\in[k,k+h]} s_j^{t+i} - \min_{i\in[k,k+h]} s_j^{t+i} \geq \epsilon\}. \tag{24}$$

The functionals g_1 and g_2 are applied to a time step t of a scenario s_j to obtain probability forecasts $P_t\left[g(s_j, k, h, \epsilon)\right]$ as follows:

$$P_t\left[g(\mathbf{S}, k, h, \epsilon)\right] = \frac{1}{S}\sum_{j=1}^{S} g(s_j, k, h, \epsilon), \tag{25}$$

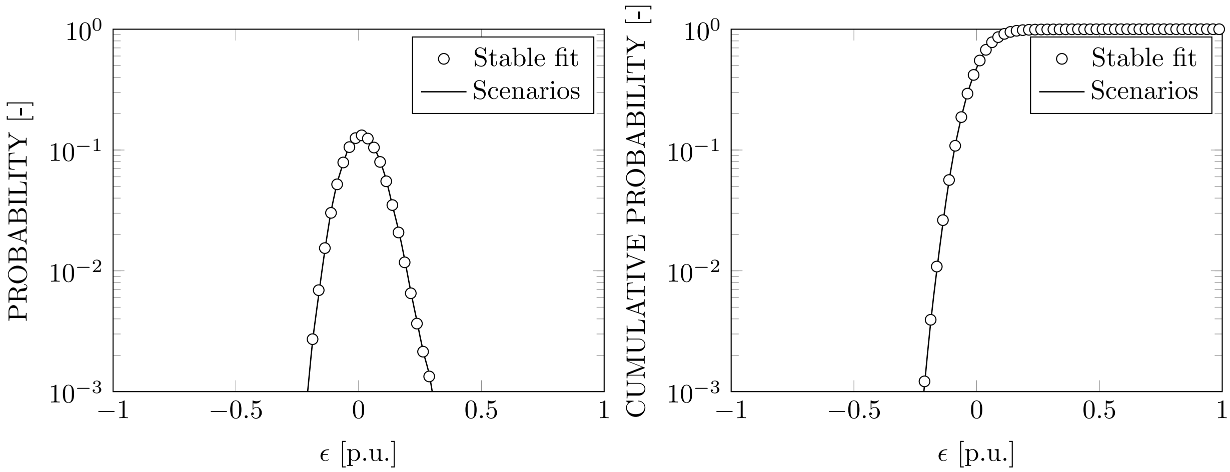
in which P_t should be understood as the probability of the event described by functional g, as predicted by the set of scenarios $\mathbf{S}$ (containing S scenarios s_j). Using this probability forecast, Brier scores Bs [5] can be defined for each event as follows by calculating the quadratic distance between the probability forecast, obtained from the set of scenarios, and the value of the functional applied to the observed wind power forecast errors at the same time step:

$$Bs = \frac{1}{T-h}\sum_{t=1}^{T-h}\left(P_t\left[g(\mathbf{S}, k, h, \epsilon)\right] - g(WPFE, k, h, \epsilon)\right), \tag{26}$$

with T the length of the scenarios considered. This Brier score Bs varies between 0 and 1, with a Bs equal to 0 indicating a perfect correspondence between the occurrence of a certain event in the set of scenarios and the observations. The Brier score can be made horizon dependent or can be aggregated as in Eq. (26).

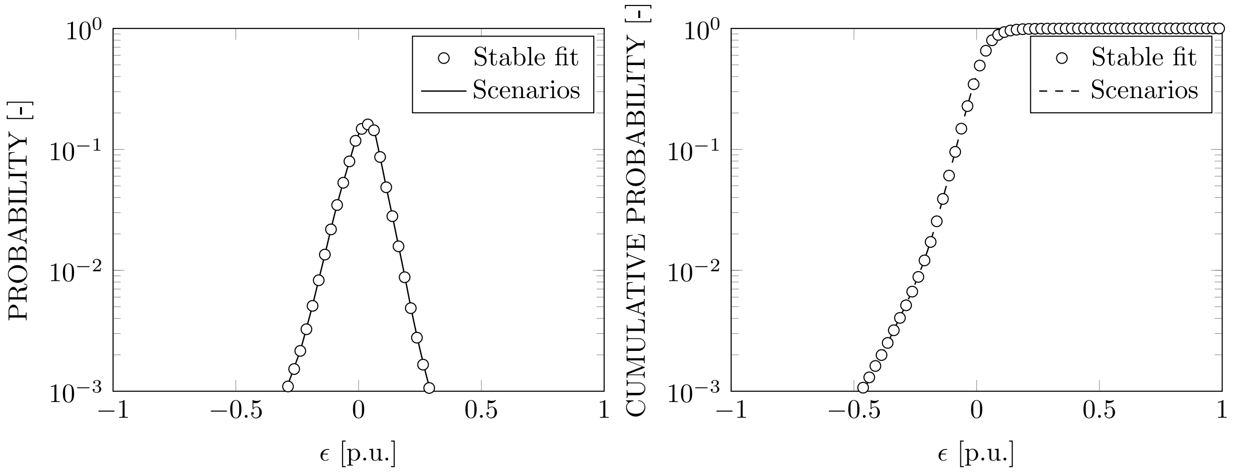
In Table 2 and 3, we have summarized the Brier scores obtained from the set of scenarios generated Belgian wind power forecast error. The values displayed are average Brier scores: we calculated the Bs for each set of scenarios, which were generated to represent the forecast error for a specific day, and averaged the resulting Brier scores over the year. The number in brackets indicates the variance on the results obtained. The Brier scores for events described by functional g_1 are all below an acceptable 20%, with the largest Bs values for events with ϵ equal to zero. Similarly, for the gradient-type events (functional g_2), Brier scores are below 20%, with the highest Brier scores for small gradients over a long period of time. In both cases, the variance on these values is small, which allows us to conclude that the presented, average Brier scores are a good representation of the Brier score for each set of scenarios. As Brier scores are low, we conclude that the set of scenarios captures the probability of occurrence of the investigate events adequately.

Conclusion In conclusion, we have shown that the technique proposed above allows to capture the behavior of the wind power forecast error with a reasonable number of scenarios. Not only the amplitude of the error, but also the variability is represented in these scenarios. Moreover, an event-based evaluation framework showed that the resulting scenarios include those events (prolonged errors, large gradients over time) that are deemed challenging for the power system, are included within the scenarios and that these occur with a reasonable probability.

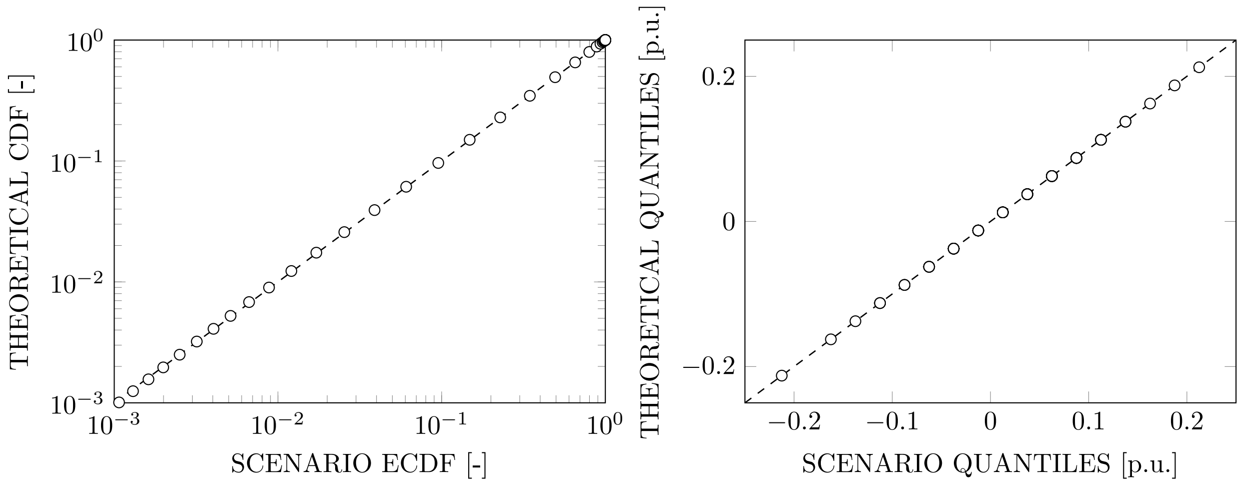


(a) Histogram and fitted distributions (pdf) (b) Cumulative histogram and fitted distributions (cdf)

Figure 7: The histogram, the empirical and 'scenarios' pdf and cdf of the wind power forecast error for the 10^{th} forecast power bin. Note the asymmetry of the data: as the 10^{th} power bin contains forecasts between 0.225 and 0.25 p.u., the error must be in the interval [-0.25 p.u.,0.75 p.u.].

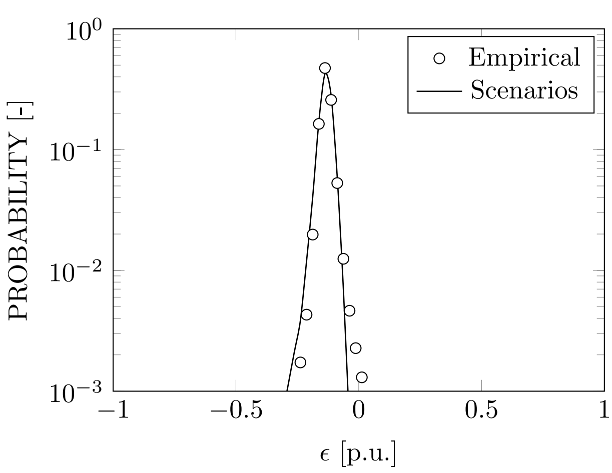


(a) Overall histogram and fitted distributions (pdf) (b) Overall cumulative histogram and fitted distributions (cdf)

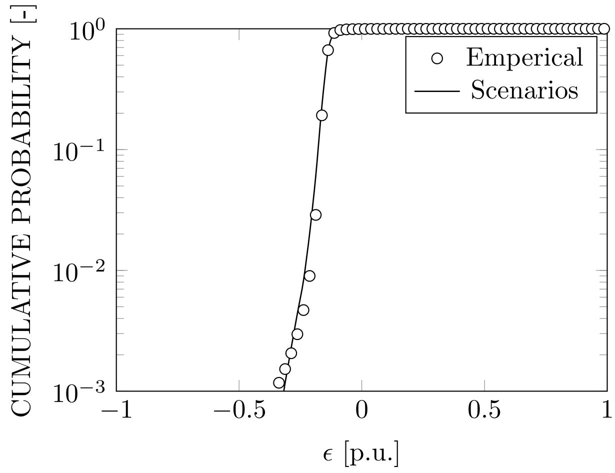


(c) Overall pp-plot (d) Overall qq-plot (percentiles)

Figure 8: The overall distribution of the generated forecast errors and the stable distribution is in perfect agreement. Subfigure (a) shows the overall probability density function (pdf). Subfigure (b) displays the overall cumulative density function (cdf). Subfigure (c) contains the pp-plot, showing the proposed (scenarios) ('theoretical') cdf versus the empirical cdf. Subfigure (d) shows the qq-plot, plotting the theoretical quanitles (percentiles) of the scenario distribution against the empircal quantiles.

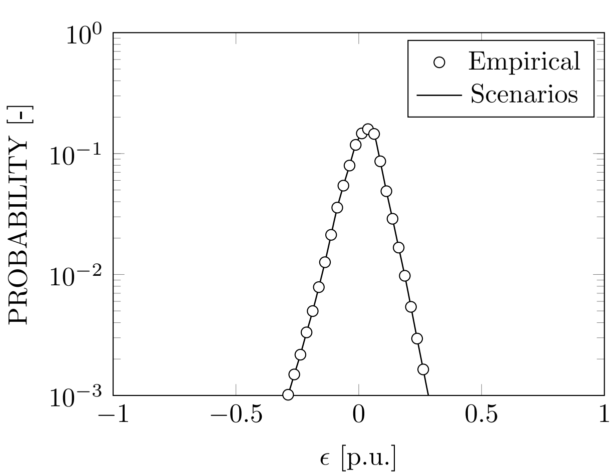


(a) Histogram and fitted distributions (pdf)

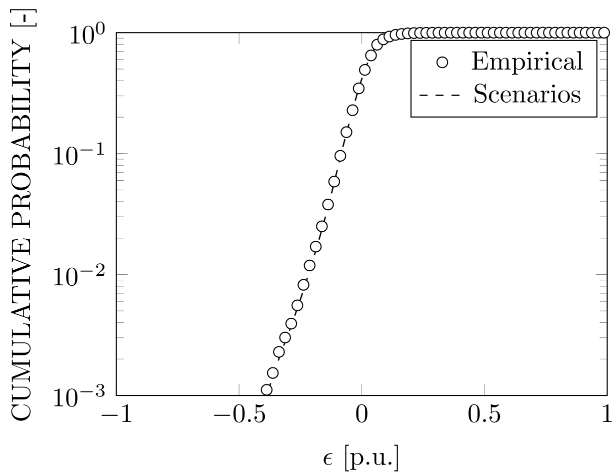


(b) Cumulative histogram and fitted distributions (cdf)

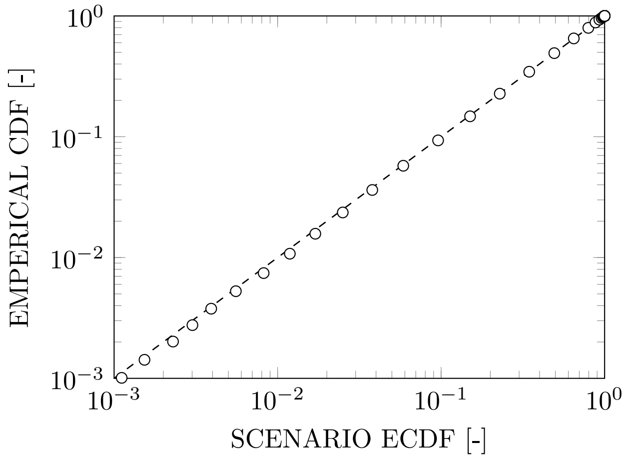
Figure 9: The histogram, the empirical and 'scenarios' pdf and cdf of the wind power forecast error for the 35$^{\text{th}}$ previous error power bin. This power bin contains errors observed subsequent to an error in the interval -0.15 and $-0.125p.u.$.



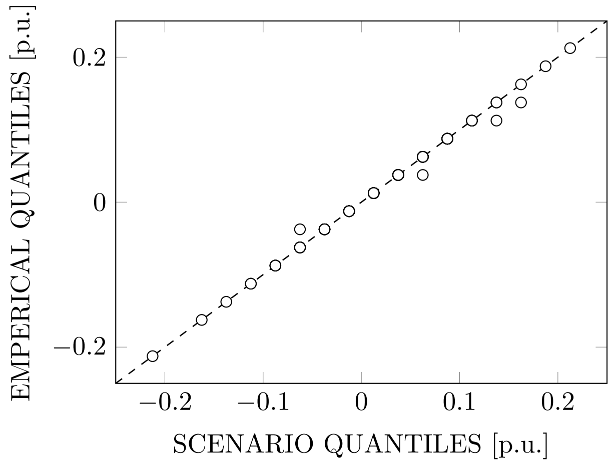
(a) Overall histogram and fitted distributions (pdf)



(b) Overall cumulative histogram and fitted distributions (cdf)



(c) Overall pp-plot



(d) Overall qq-plot (percentiles)

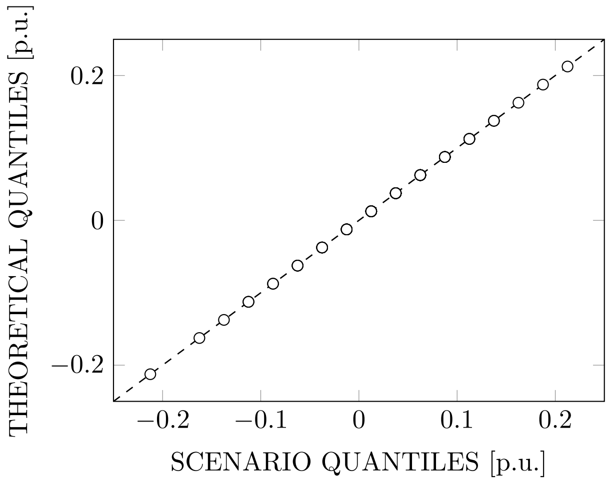
Figure 10: The overall distribution of the variability of the generated forecast errors and the empirical distribution is in good agreement. Subfigure (a) shows the overall probability density function (pdf). Subfigure (b) displays the overall cumulative density function (cdf). Subfigure (c) contains the pp-plot, showing the proposed (scenarios) ('theoretical') cdf versus the empirical cdf. Subfigure (d) shows the qq-plot, plotting the theroretical quanitles (percentiles) of the scenario distribution against the empircal quantiles.

		h [h]						
		0.5	1	2	3	4	5	6
ϵ [$p.u.$]	-0.5	0 (0)	0 (0)	0 (0)	0.0001 (0)	0 (0)	0.0001 (0)	0.0001 (0)
	-0.4	0.0007 (0.0001)	0.0011 (0.0002)	0.0015 (0.0003)	0.0018 (0.0004)	0.0022 (0.0005)	0.0025 (0.0006)	0.0027 (0.0007)
	-0.3	0.0038 (0.0015)	0.0045 (0.0017)	0.0054 (0.0021)	0.0060 (0.0023)	0.0064 (0.0023)	0.0067 (0.0023)	0.0070 (0.0023)
	-0.2	0.0138 (0.0051)	0.0165 (0.0054)	0.0190 (0.0057)	0.0208 (0.0059)	0.0221 (0.0059)	0.0231 (0.0060)	0.0238 (0.0061)
	-0.1	0.0511 (0.0112)	0.0636 (0.0122)	0.0729 (0.0130)	0.0791 (0.0134)	0.0831 (0.0133)	0.0855 (0.0131)	0.0867 (0.0127)
	0	0.1827 (0.0089)	0.1888 (0.0060)	0.1844 (0.0042)	0.1761 (0.0033)	0.1666 (0.0031)	0.1565 (0.0032)	0.1458 (0.0034)
	0.1	0.1028 (0.0129)	0.0686 (0.0102)	0.0446 (0.0072)	0.0300 (0.0049)	0.0207 (0.0034)	0.0145 (0.0023)	0.0099 (0.0014)
	0.2	0.0121 (0.0014)	0.0058 (0.0006)	0.0025 (0.0002)	0.0010 (0.0001)	0.0004 (0)	0.0001 (0)	0 (0)
	0.3	0.0013 (0.0001)	0.0006 (0)	0.0002 (0)	0.0001 (0)	0 (0)	0 (0)	0 (0)
	0.4	0.0001 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
	0.5	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)

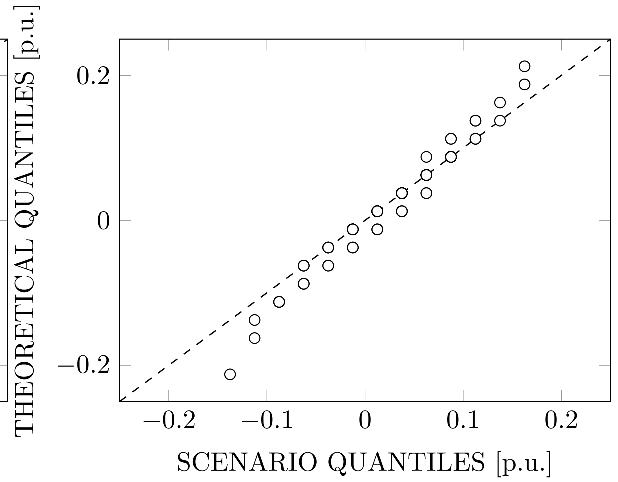
Table 2: The average Brier scores for prolonged forecast errors (functional $g_1(s_j, k, h, \epsilon) = \prod_{i=k}^{i=k+h} \mathbf{1}\{s_j^{t+i} \geq \epsilon\}$) shows that the set of scenarios include all relevant events of type g_1 and that these events occur with a reasonable probability. The value between brackets is the variance on the Brier score.

		h [h]						
		0.5	1	2	3	4	5	6
ϵ [$p.u.$]	0.1	0.0023 (0)	0.0347 (0.0027)	0.1061 (0.0123)	0.1606 (0.0183)	0.1857 (0.0168)	0.1914 (0.0126)	0.1876 (0.0091)
	0.2	0.0000 (0)	0.0029 (0.0001)	0.0106 (0.0009)	0.0230 (0.0029)	0.0378 (0.0059)	0.0534 (0.0097)	0.0689 (0.0139)
	0.3	0.0001 (0)	0.0004 (0)	0.0015 (0.0001)	0.0037 (0.0003)	0.0063 (0.0006)	0.0098 (0.0013)	0.0140 (0.0025)
	0.4	0.0001 (0)	0.0002 (0)	0.0004 (0)	0.0010 (0.0001)	0.0020 (0.0002)	0.0030 (0.0004)	0.0039 (0.0006)
	0.5	0 (0)	0.0001 (0)	0.0003 (0)	0.0005 (0.0001)	0.0008 (0.0001)	0.0010 (0.0002)	0.0012 (0.0003)

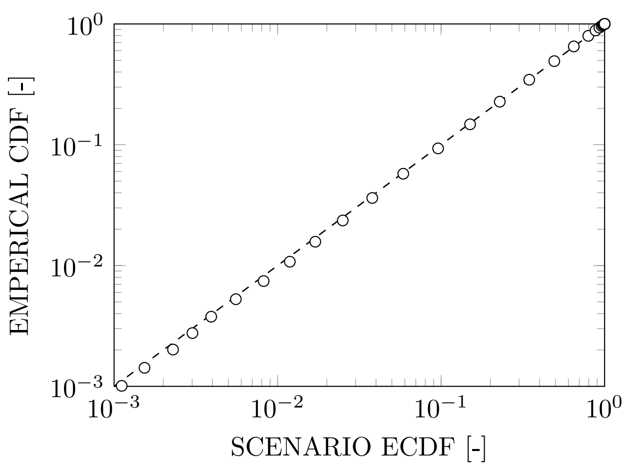
Table 3: The average Brier scores for gradients in the forecast errors (functional $gg_2(s_j, k, h, \epsilon) = \mathbf{1}\{\max_{i\in[k,k+h]} s_j^{t+i} - \min_{i\in[k,k+h]} s_j^{t+i} \geq \epsilon\}$) shows that the set of scenarios include all relevant events of type g_2 and that these events occur with a reasonable probability. The value between brackets is the variance on the Brier score.



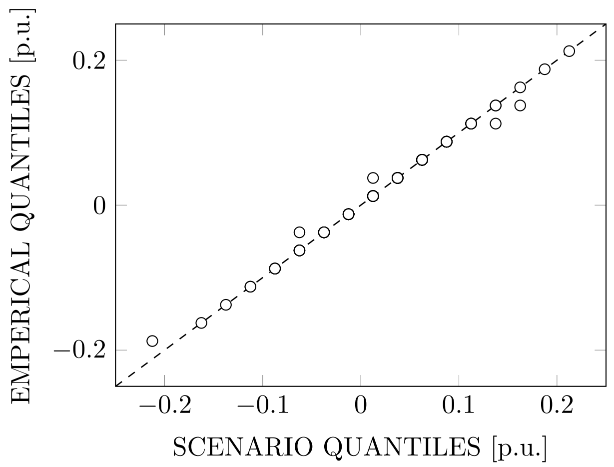
(a) Before removal improbable scenarios – probability of an error, given the forecast



(b) After removal improbable scenarios – probability of an error, given the forecast



(c) Before removal improbable scenario – probability of an error, given the previous error



(d) After removal improbable scenario – probability of an error, given the previous error

Figure 11: The effect of the removal of scenarios containing improbable events on the probability density functions of the error.

5.2 Impact of removal of improbable scenarios

As indicated above, the only relation between scenarios that is taken into account in the scenario generation technique is the covariance matrix. This covariance matrix is an estimation of the true covariance matrix, which would vary from day to day, hour to hour. As a result, the obtained scenarios may contain events – being transitions from one error to the next one – that in reality were not observed. Keeping this events in our scenarios would lead to an overestimation of the stress imposed on the power system by the wind power forecast errors. Therefore, we will remove scenarios containing such an event that has a zero probability of occurrence[8].

However, removing these scenarios – while assuming that each of the remaining scenarios remains equally probable – will distort the probability density functions (Fig. 8 and 10) of the errors in the set of scenarios. The magnitude of this distortion is examined in Fig. 11. In this Figure, the qq-plots before and after removal of the improbable scenarios are shown for the overall pdf of the error given the forecast and the overall pdf of the error given the previous error. As shown in Fig. 11, the effect is most pronounced in the probability of an error, given the forecast. Without those scenarios that contain an event with zero probability, the probability of large forecast errors is slightly underestimated. For the variability of the error, the effect is negligible.

Although this might affect the results to some extend, we have opted to proceed with the corrected set of scenarios, as the error we might trigger by overestimating the variability of the error is much larger.

[8]The probability of an error given the previous error is calculated from the empirical distribution used as a benchmark in Fig. 7 and 10.

5.3 Scenario reduction & the stability of the stochastic program

To illustrate the added value of the proposed scenario reduction method, we will study the **stability** and **bias** of the resulting solution of the corresponding stochastic program. We will do so for the stochastic unit commitment model with which one can study the impact of wind power forecast errors on the Belgian power system. A full description of the model, data and assumptions can be found online [11][9]. As stability analyses are computationally demanding, it is impossible to do so for each day of the year. In this case, we are studying the Belgian power system with an assumed wind power penetration of 30% or 10 GW of installed wind capacity, of which we will select the day with the residual demand closest to the yearly average. The scenarios generated for this day are illustrated in Fig. 6.

Stability analysis The stability analysis discussed below is based on Kaut et al. [7], as discussed above. In-sample stability (i.e. stability of the first stage objective function of the stochastic program) will be claimed if the objective value of the stochastic program does not change too much if more scenarios are considered. Out-of-sample stability is reached if the objective value of the stochastic program considering the full set of scenarios, obtained with fixed first stage variables, does not change too much with the addition of more scenarios to the stochastic program to obtain the first stage variable and is close to the first stage objective value. For this particular stochastic program, one can expect the following trends. For a low number of scenarios, the stochastic unit commitment model will decide that there is relatively little need for online capacity – it has only knowledge of a limited number of possible events that could occur. As a result, the first stage objective value will be relatively low. However, when reevaluating this solution on a large set of scenarios, the second stage objective value will be extremely high, as the proposed schedule will not be adequate to avoid loss of load. If more scenarios are added, more online capacity will be scheduled, leading to higher first stage objective values. As a result, less loss of load will be incured during the second stage optimization, leading to reduced second stage objective values. As the number of scenarios increases, the first and second stage objective values converge. For a detailed discussion, see [1].

In Fig. 12, the results of the stability analysis are shown. On the left ('original'), we display the results from the stochastic program considering the original fast forward algorithm [4], in which the probability distance between two scenarios is calculated based on the two-norm of error. On the right ('modified'), the evolution towards stability is shown for the proposed scenario reduction method, based on the fast forward algorithm, but with the calculation of the probability distance between two scenarios based on the difference in objective function of two equivalent deterministic one-scenario problems. As is evident from Fig. 12, stability is not reached in the case of the original fast forward algorithm. Although the first stage objective value (solid lines) reaches a seemingly steady value with 30 scenarios and more (in-sample stability), the second stage objective value does not converge (no out-of-sample stability). Moreover, the difference between the first stage and second stage objective values remains relatively large: the difference amounts to more than 5% when considering 50 scenarios in the first stage optimization. When using the modified scenario reduction algorithm, in- and out-of-sample stability can be claimed when considering 15 scenarios or more. The first stage objective value remains at the same level as of 15 scenarios, while reallocation of the scheduled capacity further decreases the second stage objective value to levels less than 1% above the first stage objective value. Note that this lies within the optimality gap of this optimization (1%).

The underlying reason for this difference in stability was already briefly discussed in Section 4. As illustrated in the example, the 'original' scenario reduction only looks at the amplitude of the scenario itself, but not at the impact of that scenario on the result of the stochastic program. This might be due the variability of the scenario, a severe negative forecast error, etc., all of which are reflected in the cost of a one-scenario equivalent of the stochastic program for each scenario. To further illustrate this point, Fig. 13 displays the two times two sets of scenarios with the same cardinality, once obtained with the 'original' scenario reduction technique, once with the modified version. By comparing the sets of scenarios of the same cardinality, it is clear that the 'modified' scenario reduction technique succeeds better in methodologically selecting a wide range of scenarios, with a focus on those that are challenging for the power system (i.e. large negative forecast errors). Because generation costs increase non-linearly with the demand – i.e. peak power units are more expensive per MWh/h than nuclear power plants – the 'cost' of scenarios with severe or prolonged negative forecast errors will be further from the cost of the forecast scenario than that of the scenarios with positive forecast errors. As a result, this technique will have the tenancy to select more scenarios that are costly (and thus in general have large negative forecast errors) compared to the inexpensive scenarios. This is illustrated in Fig. 13. If 15 scenarios are retained, the difference between the proposed and the original *fast forward scenario reduction* method is clear. The modified scenario reduction technique succeeds better in capturing the range of objective functions of the deterministic one-scenario equivalent unit commitment problems, but is somewhat biased towards the scenarios with a high operational cost (i.e. negative forecast errors). However, for 30 scenarios, the differences are much more subtle between the two sets of scenarios. However, the objective function of the corresponding stochastic programs is

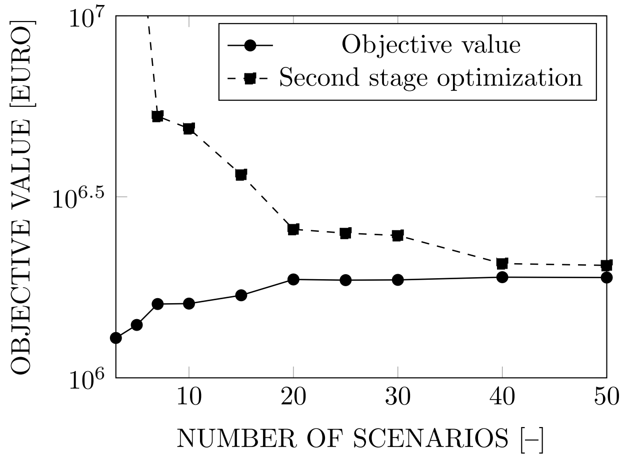
[9]Note that in this paper, the indirect start-up costs and the ramping costs were not taken into account.

significantly different (Fig 12). The difference between the two scenario trees lies in the probability assigned to each scenario. As outlined in Section 4, the probability of the deleted scenarios is reassigned to the retained scenarios based on the Kantorovich distance between the scenarios. In the original fast forward algorithm, this leads to an underestimation of the probability of extreme scenarios, which leads to sub-optimal power plant scheduling and load shedding.

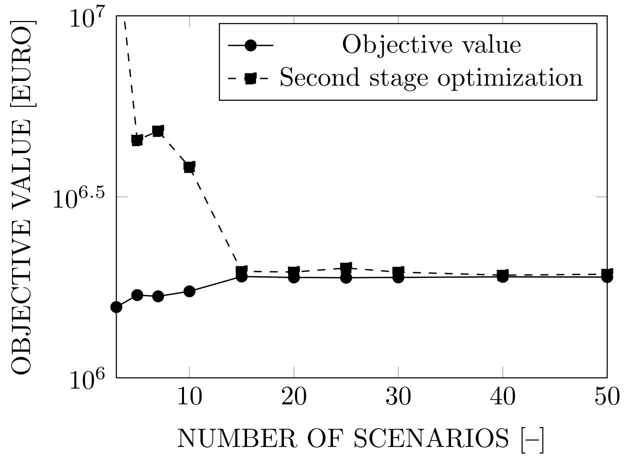
Bias – results to be updated

Computational burden of the scenario reduction algorithm – results to be updated

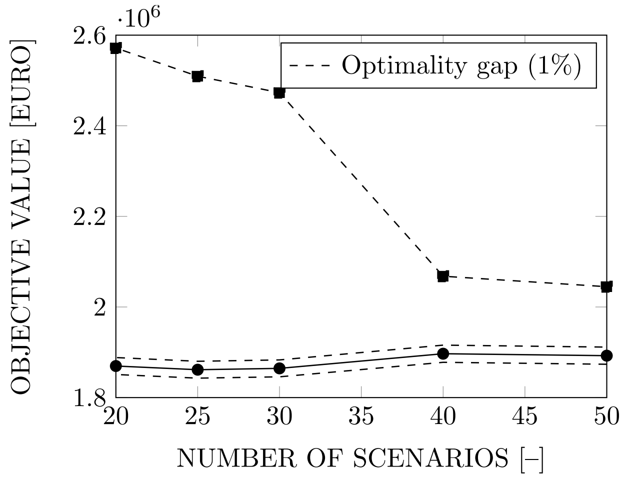
Conclusion In this section, we have shown that the fast forward scenario reduction algorithm can be used to select the relevant wind power forecast error scenarios if one uses the (difference in) operational cost, obtained from the equivalent one-scenario deterministic unit commitment problem, to distinguish between the scenarios. This allows us to reach stability in the solution of the stochastic program with relatively little scenarios. The reason for this difference in performance lies in the non-linear increase of electricity generation costs with the demand, leading to an emphasis on negative forecast errors, and an adequate assignment of probability to each scenario, weighing the representation of the 'average' and 'extreme' scenarios.



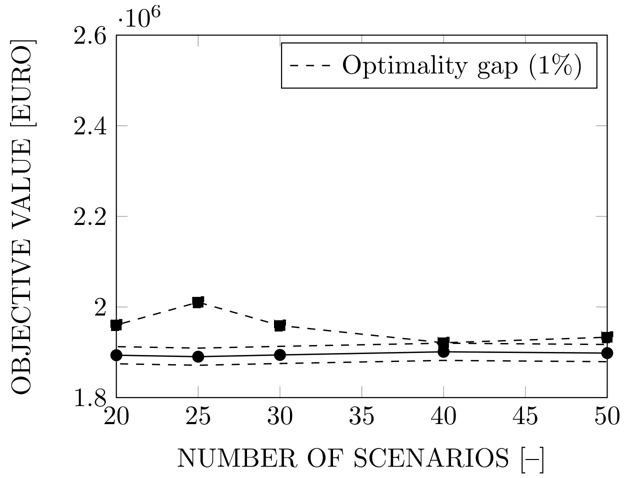
(a) 'Original' scenario reduction technique



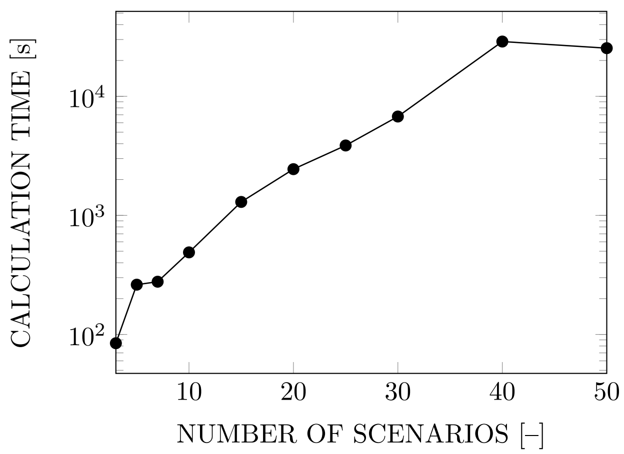
(b) 'Modified' scenario reduction technique



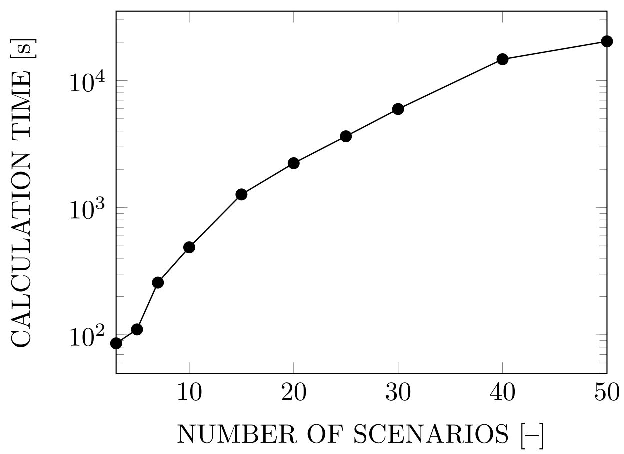
(c) 'Original' scenario reduction technique



(d) 'Modified' scenario reduction technique

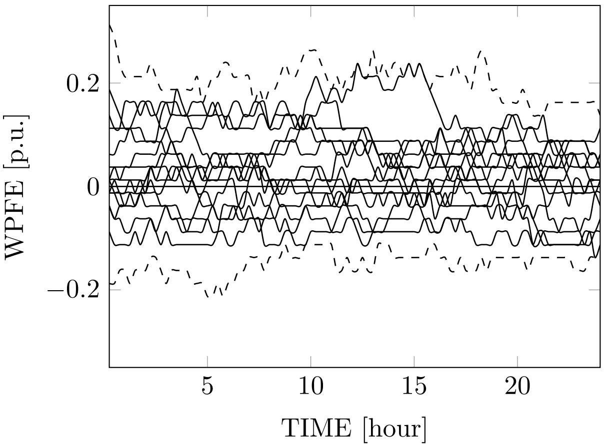


(e) 'Original' scenario reduction technique

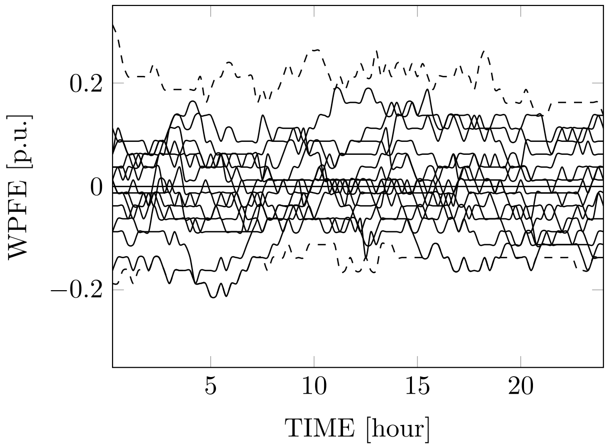


(f) 'Modified' scenario reduction technique

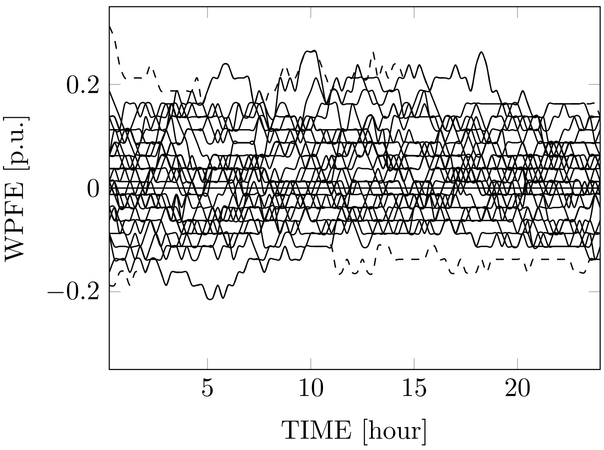
Figure 12: With the 'Modified' scenario reduction technique, stability is reached as of 15 scenarios. The 'original' scenario reduction technique does may lead to in-sample stability, but out-of-sample stability is not reached. The calculation times of the corresponding stochastic programs remains unaffected.



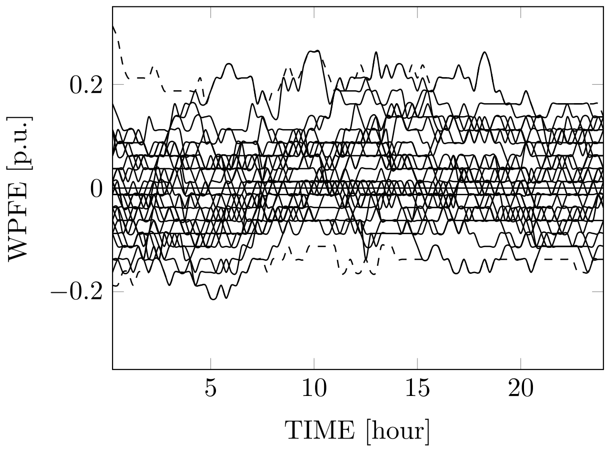
(a) 'Original' scenario reduction technique – 15 scenarios



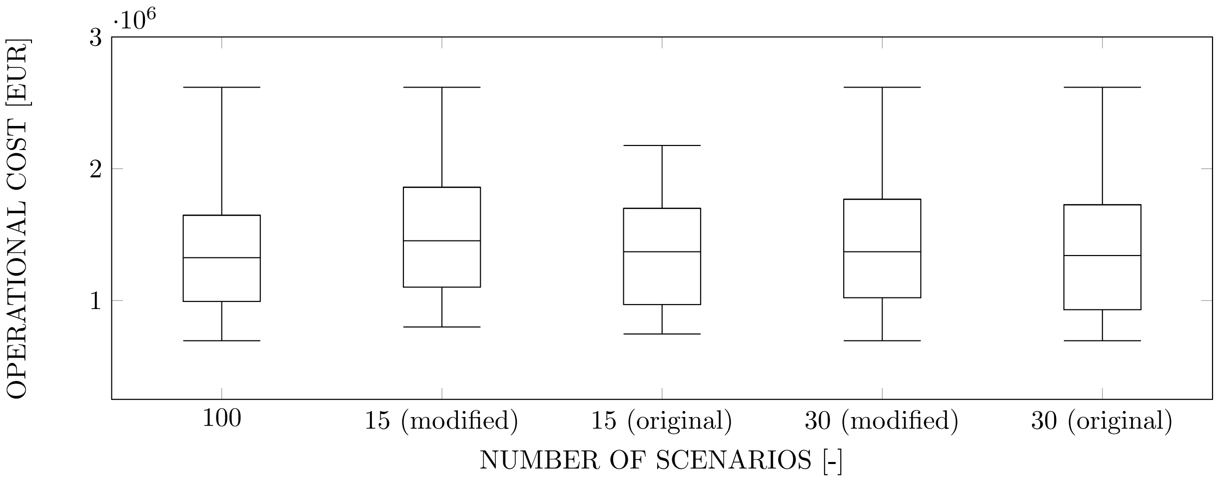
(b) 'Modified' scenario reduction technique – 15 scenarios



(c) 'Original' scenario reduction technique – 30 scenarios



(d) 'Modified' scenario reduction technique – 30 scenarios



(e) The distribution of the (selected) deterministic one-scenario equivalent unit commitment problems.

Figure 13: The modified *fast forward scenario reduction algorithm* is capable of selecting a limited set of scenarios that captures the cost distribution of the stochastic problem (Subfigure (e)). Due to the non-linear increase of electricity generation costs with demand, the modified method puts more emphasis on the negative forecast errors (Subfigure (a)-(b)). Although there is no clear difference between the selected scenarios for sets of higher cardinality (Subfigure (c)-(d)), only the modified method leads to stable results, as only this method allows for an adequate redistribution of the probability to the selected scenarios.

6 Conclusion

Concluding remarks goes here.

Future work Beyond this work, one needs to

- look at the effect of the assumed probability density functions;
- note that scenarios are a part of the model, and thus each model and each uncertain parameter will need his own, adapted scenario generation and or reduction technique.

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